NATIONAL UNIVERSITY OF HO CHI MINH CITY

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

FACULTY OF MECHANICAL ENGINEERING

MECHATRONICAL PROJECT
(ME4071)

DESIGN LINE FOLLOWING ROBOT THAT DETECTS AND AVOID THE OBSTACLE PATH

Semester: 261/ **Year:** 2026 – 2027

GVHD: PGS. TS. Nguyễn Tấn Tiến

SVTH: Đoàn Trần Phương Đông **MSSV:** 2310755

Đặng Văn Đức

Bùi Huy Hoàng

Đỗ Lê Anh Khoa

Ho Chi Minh City, 2026

TOPIC 3

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PROJECT TOPIC

General requirement

Groups will select one of the four given topics. Based on the technical requirements of the topic, students will propose technical alternatives through general analysis and research. These alternatives must then be analyzed and compared against specific evaluation criteria to select the most optimal solution. From this chosen solution, the detailed design of the mechatronic system will be carried out. The design must be verified through both simulation and experiment (a physical prototype).

The topics require designing a robot to perform various line-following tasks on an arena. The characteristics of the arena are described as follows:

(i) The arena background is white.

(ii) The guideline is black.

(iii) Line width: 26 mm

(iv) Planar movement

(v) Minimum moving speed: 0.1 m/s

(vi) Line tracking error (measured from either the left or right edge of the line): ±3 mm.

(vii) Stopping position error at the end of the line: ±5 mm.

Note: Students must self-define the tracking point on the robot and use it to evaluate the aforementioned errors.

Topic 3: Line-following robot with obstacle detection.

Students are required to design a robot capable of detecting an obstructed route and selecting the appropriate alternative path. The robot will operate on the arena shown in Figure 3.

The operational requirements of the system are as follows:

- The robot moves from the "Start" position to one of the two "Finish" positions.
- An obstacle will be randomly placed in one of the two "Obstacle zones". The robot must detect the blocked path and travel along the remaining clear path.

Note: The obstacle is a cylinder with a cross-sectional diameter of 50 mm.

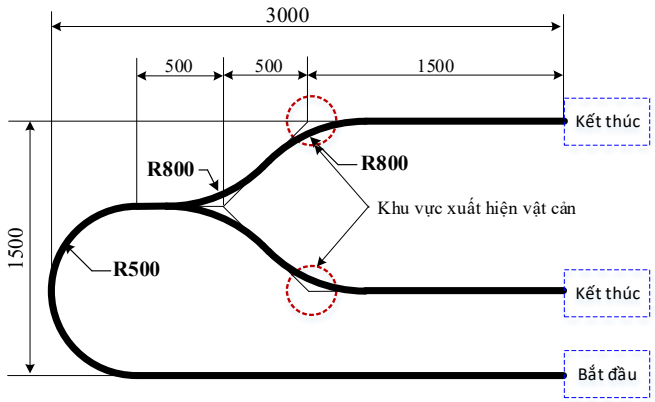


Figure. Area diagram for line following robot with obstacle detection

ACKNOWLEGEMENTS

TOPIC 3

GANTT CHART

Tasks	Responsibility	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
I. OVERVIEW	Team															
Introduction & research about the topic																
Research on available models																
Parameters selection & Approach																
Overview adjustment																
II. DESIGN																
1. Mechanical	Khoa															
Calculate and select motor																
Design the frame																
Draw 3D drawing & adjust parameters																
Draw 2D drawing & adjust parameters																
2. Electrical	Đông															
Build schematic diagram of electrical system																
Select components of electrical system																
Design PCB																
Testing PCB principle and sensors																
Read data from sensors and calibration																
Adjust parameters																
3. Control	Hoàng															
Model the kinematics for the robot																
Determine transfer function and design PID controller of motor																
Design line-following controller																
Select obstacle avoidance path																
Adjust parameters																
4. Simulation	Đức															
Programming																
Define simulation requirements																
Specify parameters																
Simulation & evaluate data																
III. Manufacturing	Team															
Build PCB and wiring vehicle system																
Fabricate robot body & assembly mechanical components																
Assembly the robot																
IV. Experiment & Tuning																
V. Completion																
Complete report & drawings																
VI. Project defense																

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CHAPTER 1. OVERVIEW

1.1. Necessity

In modern manufacturing, warehousing, and logistics systems, the demand for automating the transport of goods, materials, and components between workstations has been steadily increasing. Manual transport depends heavily on human labor, makes it difficult to maintain high accuracy and repeatability, and increases both operating time and cost. As a result, mobile robots and Automated Guided Vehicles (AGVs) are being increasingly deployed to carry out transport tasks along predefined routes.

According to the World Robotics 2025 report published by the International Federation of Robotics (IFR), approximately 102,900 professional service robots in the transportation and logistics segment were sold in 2024, representing a 14% increase compared with the previous year. This category accounted for more than half of all professional service robots sold and consists mainly of mobile robots used for transport and goods handling[1].

Among the various guidance methods for mobile robots, line following is a relatively simple, stable, and low-cost solution. The robot uses a sensor array to determine the relative position of the line and controls two motors to follow a predefined trajectory. Compared with more complex localization approaches based on LiDAR, cameras, or environment mapping, a line-following system requires simpler hardware and control algorithms, making it well suited to environments with fixed transport routes such as factories, warehouses, and internal material-handling systems.

However, real-world operating environments are not entirely static. Along the travel route, people, goods, equipment, vehicles, or other robots may appear and obstruct the path. If the robot merely follows the line without any obstacle-recognition capability, it may collide with these obstacles or be forced to halt, thereby reducing the safety, flexibility, and continuous availability of the transport system.

For this reason, integrating obstacle-detection capability into a line-following robot is essential. In a guideway system with multiple branches, the robot must be able to detect when its current route is blocked and select an alternative route to continue operating, rather than stopping at the obstruction. This capability helps reduce the risk of collision, minimize interruption time, and enhance the overall autonomy of the system.

Based on these requirements, the study and development of a line-following robot capable of obstacle detection and automatic alternative-route selection holds significant practical value in improving the level of automation of transport systems. This work addresses both the problem of stable trajectory tracking and movement, and the extended capability of environmental perception and route selection, providing a foundation for AGV models applied in factories, warehouses, and automated logistics systems.

1.2. Problem statement

A line-following robot is a type of autonomous mobile robot capable of moving along a predefined trajectory using a sensor system to detect the guideline. During operation, the robot

must ensure the ability to recognize the line, control its motion stably, and move accurately along the required route.

However, obstacles may appear on the path. For conventional line-following robots, upon encountering an obstacle, they might have to stop or wait for it to be removed, causing interruptions in the transportation process. Specifically, for systems with multiple branches, the robot needs the capability to identify the blocked route and select an alternative path.

Therefore, the objective of this project is to develop a robot capable of detecting and tracking a line, while simultaneously detecting obstacles and selecting an appropriate route of travel. The robot utilizes line-tracking sensors to maintain its trajectory and distance sensors to detect obstacles at the branching area.

In the project model, the robot departs from the "Start" position and follows the line to the branching area. One of the two routes leading to the "Finish" position will contain an obstacle. The robot must identify the blocked route, choose the remaining clear route, and continue moving to the destination without any collisions.Để thực hiện yêu cầu trên, hệ thống cần giải quyết các vấn đề chính:

To fulfill the above requirements, the system needs to resolve the following key issues:

- Line detection and tracking: Identifying the position of the guideline and controlling the speeds of the two motors so the robot moves stably.
- Obstacle detection: Recognizing objects on the path to give the robot sufficient time to make a decision.
- Branching area recognition: Determining the exact moment when a route selection must be made.
- Route selection and turning: Identifying the blocked branch, selecting the remaining clear branch, and executing the turn without losing the line.
- Maintaining stable motion: After passing the branching area, the robot must continue tracking the line and move to the finish position.đến vị trí kết thúc

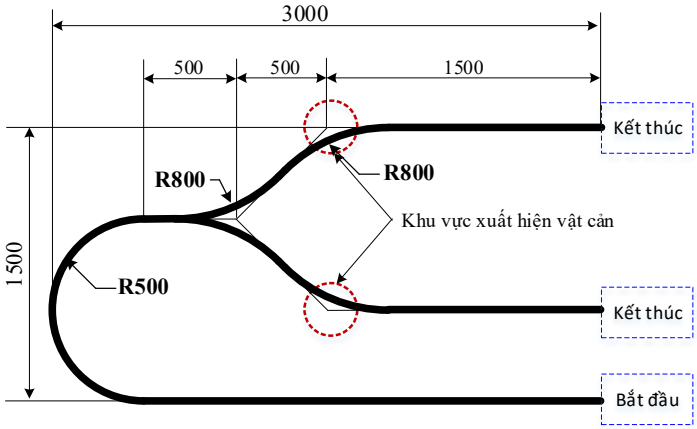


Figure 1.1. Arena layout and the route selection problem of the line-following robot

1.3. Pololu 3pi+ 32U4

The Pololu 3pi+ 32U4 is a compact, user-programmable mobile robot developed by Pololu Corporation for educational and autonomous robotics applications. With an overall diameter of approximately 97 mm, the robot integrates an ATmega32U4 microcontroller, two independently driven DC gearmotors, quadrature encoders, an inertial measurement unit, and five downward-facing infrared reflectance sensors. These reflectance sensors allow the robot to distinguish between light and dark surfaces and can therefore be used for line-following applications. The robot employs a differential-drive locomotion configuration, in which its moving direction is controlled by varying the speeds of the left and right drive wheels[2].

1.3.1. Mechanical system

1.3.1.1. Mechanical structure and motion principle

a. Mechanical structure

The mechanical structure of the Pololu 3pi+ 32U4 consists primarily of two independently driven wheels, two DC gearmotors, a chassis, and a passive ball caster. The five line sensors are located underneath the front edge of the control board and face downward toward the track surface. The chassis also incorporates the battery compartment and provides mounting positions for the motors and control board [2].

A simplified mechanical principle diagram can be drawn as follows:

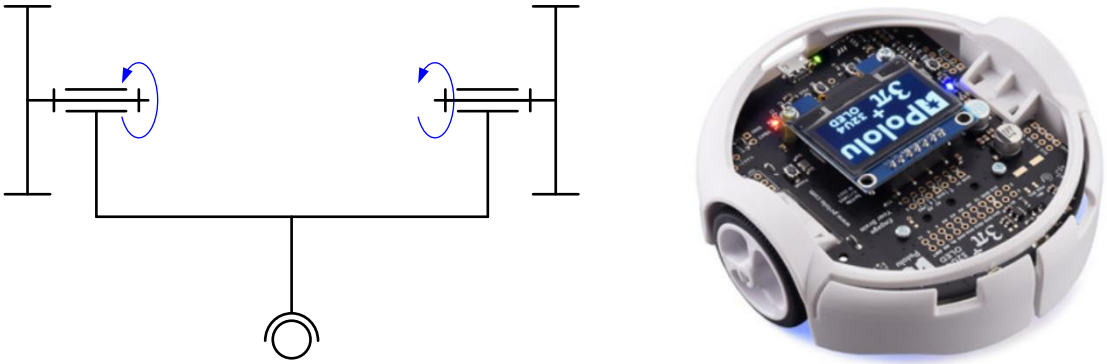


Figure 1.2. Line following robot Pololu 3pi+ 32U4

b. Number and function of motors

The robot uses two DC gearmotors with an extended rear shaft for encoder installation. The left motor drives the left wheel, while the right motor drives the right wheel. Both motors are independently controllable in speed and rotational direction. In addition to generating forward and backward motion, the two motors are also responsible for steering the robot by creating a difference between the left and right wheel velocities.

For the Standard Edition, Pololu uses 30:1 MP 6 V Micro Metal Gearmotors[3]. Other versions use different gear ratios, but the fundamental mechanical configuration remains unchanged.



Figure 1.3. The 30:1 MP 6V Micro Metal Gearmotors

Table 1.1. Parameters of 30:1 MP 6V Micro Metal Gearmotors

Feature	Value	Unit
Rated voltage	6	V
Gear ratio	29.86 : 1	–
Stall current	0.67	A
No-load current	70	mA
No-load speed	720	RPM
Extrapolated Stall Torque	0.45	Kg. cm

c. Steering method

The Pololu 3pi+ 32U4 does not use a separate steering mechanism such as an Ackermann linkage or steering servo. Instead, it uses a differential drive locomotion principle, in which the robot changes direction by independently varying the rotational velocities of the two drive wheels.

Let v_L and v_R denote the linear velocities of the left and right wheels, respectively. The basic motion states can be described as:

$v_L = v_R > 0$, the robot moves straight forward.

$v_L < v_R$, the robot turns left.

$v_L > v_R$, the robot turns right.

$v_L \approx -v_R$, the robot rotates approximately in place about a point between the two drive wheels.

This locomotion principle is particularly suitable for line-following applications because the steering correction can be directly converted into a difference between the commanded left- and right-wheel speeds.

d. Line sensor arrangement

The robot has five downward facing reflectance sensors, numbered from s_1 to s_5 from the robot's left side to its right side. They are arranged along the front edge on the underside of the control board, forming a transverse sensor array in front of the main drive wheel region.[2]

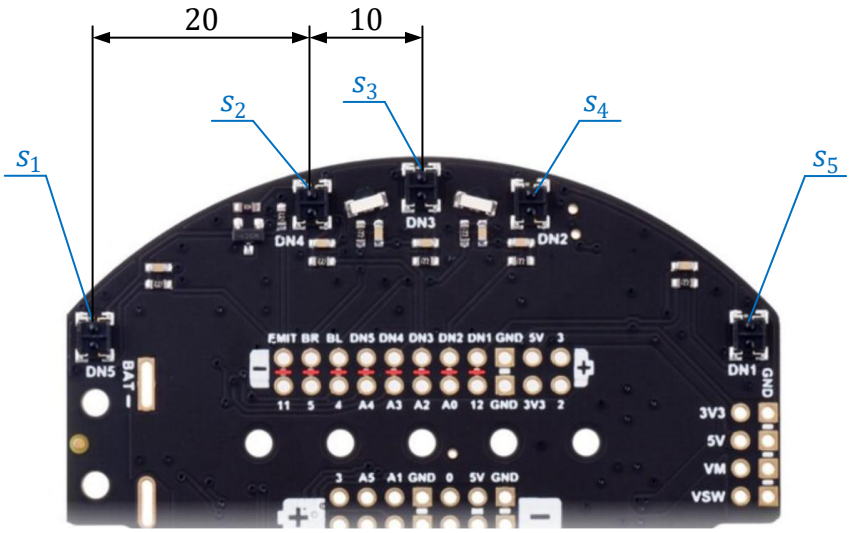


Figure 1.4. Bottom view of the 3pi+ 32U4 Control Board

1.3.1.2. Summary mechanical parameters of robot Pololu 3pi+ 32U4

Table 1.2. Summary mechanical parameters of robot Pololu 3pi+ 32U4

Feature	Description
Overall dimensions	97 × 96 × 36mm
Mechanical Structure	Two wheeled drive and a caster wheel
Number of drive motors	2
Motor	30:1 MP 6 V Micro Metal Gearmotor
Steering mechanism	Diffrential drive
Wheel diameter	32mm
Passive support diameter wheel	12.7mm
Max speed	1.5m/s
Line sensor arangement	5 downward facing reflectance sensors, along the front edge on the underside of the control board

1.3.2. Electrical system

The electrical system of the Pololu 3pi+ 32U4 integrates the sensing, processing, actuation, feedback, and power-supply subsystems on a single control board. The main electrical

components include a central microcontroller, two motor drivers, five line sensors, two wheel encoders, two bump sensors, an inertial measurement unit, and the power-regulation circuits.

1.3.2.1. Electrical system architecture

The electrical system consists of a central microcontroller, line sensors, obstacle-detection sensors, an inertial sensing unit, wheel encoders, motor drivers, two drive motors, and the power-supply circuits. The connections are divided into power lines, which supply electrical energy to the system, and signal lines, which transfer sensing and control information between the functional blocks.

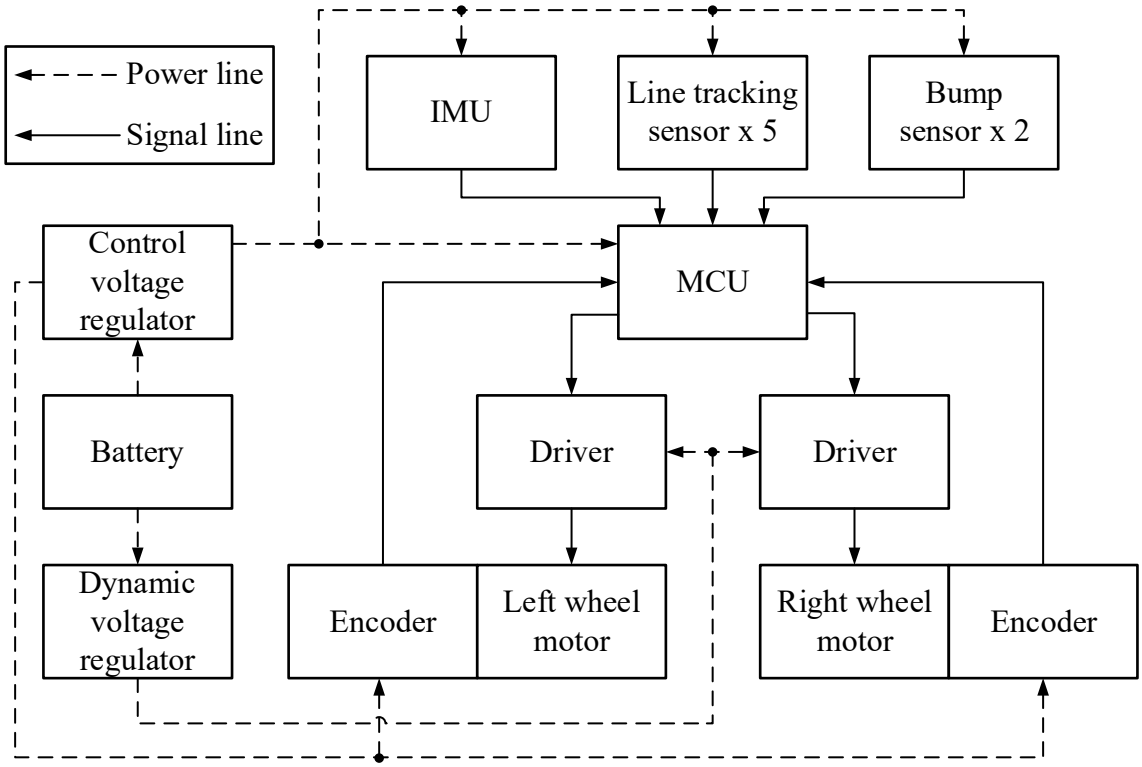


Figure 1.5. Pololu 3pi+ 32U4's electrical system architecture block diagram

During operation, the sensors and encoders send information to the microcontroller. The microcontroller processes these signals and generates control commands for the left and right motor drivers, which then drive the corresponding motors. The power subsystem supplies the control, sensing, and actuation circuits through separate voltage-regulation stages to meet their different voltage and power requirements.

1.3.2.2. Microcontroller

The central processing unit of the robot is the ATmega32U4, a high-performance 8-bit AVR microcontroller manufactured by Microchip. The MCU operates at a clock frequency of 16 MHz. It functions on a standard logic voltage of 5V.

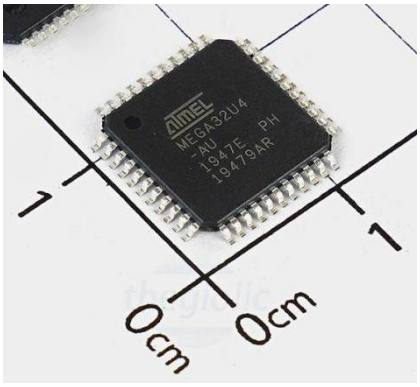


Figure 1.6. IC ATmega32U4 AVR

To interface with the onboard hardware, the ATmega32U4 relies on a specific subset of its internal peripherals tailored to the robot's subsystems:

The 16-bit Timer1 peripheral is strictly dedicated to generating precise Pulse Width Modulation (PWM) signals for the motor drivers to dictate wheel speed.

The external interrupt and pin-change interrupt pins are mapped directly to the quadrature wheel encoders.

The Two-Wire Interface (I2C) is utilized exclusively to establish high-speed communication with the 9-axis IMU.

Standard digital pins are used to control the rotation direction (DIR) of the motor drivers, toggle the infrared LEDs (EMIT) for the line and bump sensors, and drive the SPI/I2C-based OLED or LCD display for user feedback.

1.3.2.3. Motor driver and actuator

a. Motor driver

To control the robot's drive system, two DRV8838 motor driver circuits are integrated into the main board. The motor drivers are powered by a 8V voltage regulator.

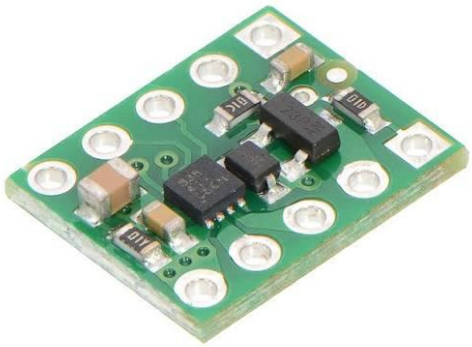


Figure 1.7. DRV8838 motor driver DRV8838 IC

The primary role of the DRV8838 drivers is to safely bridge the low-power logic of the microcontroller with the high-current demands of the actuators, while also offering overcurrent protection that automatically shuts down the 8V regulator if the motor shaft is stalled.

Regarding the signal flow, the inputs to the drivers consist of four digital control signals from the ATmega32U4 MCU. Specifically, the drivers receive Pulse Width Modulation (PWM) signals to regulate speed, alongside logic signals to dictate the rotation direction.

b. Actuator

Serving as the mechanical actuators, the system utilizes two miniature metal gear motors, specifically the 30:1 MP 6V configuration. These motors are uniquely designed with an extended rear shaft to accommodate the installation of magnetic quadrature encoders.

Functionally, each motor is responsible for independently driving one of the robot's two wheels. This independent dual-motor operation forms the core of the robot's differential drive mechanism.

1.3.2.4. Line Sensing System

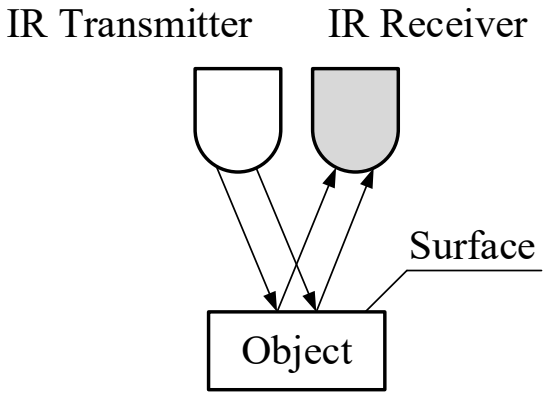


Figure 1.8. Operating principle of infrared reflective sensor

The line-following system uses five infrared reflectance sensors facing downward toward the ground. Each sensor consists of an infrared LED paired with a phototransistor to detect reflected light. During operation, the microcontroller first drives the sensor output to a high voltage level and then measures the time required for the signal voltage to decay. Dark surfaces absorb more infrared light and reflect less energy, resulting in a longer discharge time.

To ensure accurate operation and reduce interference from ambient infrared light, the system performs calibration procedures to determine the reflectance characteristics of both light and dark surfaces. The infrared LEDs of all five sensors are controlled by a single signal pin, allowing the robot to turn the LEDs on or off programmatically when required.

The five IR reflectance sensors communicate directly with the MCU through general-purpose I/O pins. These pins are first configured as outputs to charge the sensors and are then switched to input mode. The MCU uses internal timers to measure the signal decay time through the same pins, allowing the reflectance of the surface beneath each sensor to be determined.

1.3.2.5. Obstacle detection and avoidance system

The default obstacle detection system of the 3pi+ 32U4 robot does not use ultrasonic or laser sensors. Instead, it uses two bump sensors positioned at the front of the robot.

Operating principle: Although they function as mechanical bump sensors, they are actually IR reflectance sensors. Rather than measuring the reflectance of the ground, these sensors detect changes in the intensity of reflected infrared light when the front skirt flaps are pressed inward as a result of physical contact with an obstacle.

Infrared emitter control: The infrared LEDs of the two bump sensors share a common control signal, EMIT. The microcontroller must pull the EMIT pin LOW to activate the infrared LEDs. Each sensor output is protected by a $220\ \Omega$ resistor to limit current and prevent short-circuit conditions.

Environmental characteristics and calibration: Because these sensors rely on infrared light, they are sensitive to ambient lighting conditions, such as sunlight and incandescent lamps. To ensure reliable obstacle detection, the robot performs a calibration procedure by measuring baseline readings while the skirt flaps are in their normal, unpressed position. A collision is then detected by evaluating the difference between the current sensor readings and these baseline values.

Inertial Measurement Unit (IMU) In addition to the optical and contact sensors, the robot is equipped with a full 9-axis Inertial Measurement Unit (IMU), which includes a 3-axis accelerometer, a 3-axis gyroscope, and a 3-axis magnetometer.

Operating Principle (Input): The IMU physically detects linear acceleration, angular velocity, and the earth's magnetic field intensities acting upon the robot's body.

MCU Interface (Output): These physical variations are digitized by the IMU's internal ADCs and transmitted to the ATmega32U4 microcontroller via the I2C communication protocol.

Application: By reading these I2C signals, the MCU can perform sensor fusion to calculate the robot's real-time tilt, heading, and orientation, enabling complex navigation tasks that standard wheel encoders cannot fully resolve.

1.3.2.6. Feedback and additional sensors

a. Quadrature encoders

Each drive motor is equipped with a quadrature encoder consisting of a magnetic disc attached to the extended motor shaft and two Hall-effect sensors mounted on the control board. The encoders provide 12 counts per motor-shaft revolution when both edges of both channels are counted. With the Standard Edition's 29.86:1 gearbox, this corresponds to approximately 358.3 counts per wheel revolution.

To reduce the number of interrupt inputs required, the control board combines the two encoder channels using an XOR circuit. The XOR output and Channel B are read by the microcontroller and used to determine the wheel rotation count and direction.

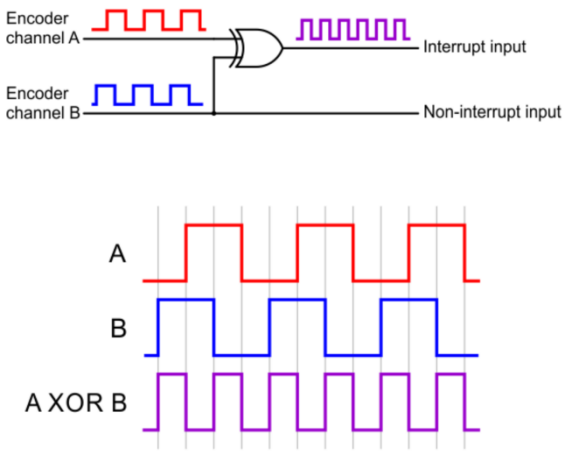


Figure 1.9. Signal timing diagram illustrating the combination of Encoder

b. Bump sensors

The robot has two forward-facing bump sensors. Although they act as contact sensors, their sensing principle is based on infrared reflectance. When the front bumper skirt is pressed inward by physical contact, the reflected infrared intensity changes and can be detected by the microcontroller.

c. Inertial measurement unit

The robot includes a 9-axis inertial sensing system consisting of an LSM6DS33 3-axis accelerometer and gyroscope and an LIS3MDL 3-axis magnetometer. These devices operate at 3.3 V and communicate with the 5 V ATmega32U4 through an I²C interface and onboard level shifters.

Together, they can be used to track the robot's trajectory, execute precise turns, or detect physical impacts. Additionally, the 3-axis magnetometer functions as a compass by measuring the Earth's magnetic field to determine the robot's heading.

The inertial data can be used for motion and orientation estimation. However, these sensors are not required by the official Pololu line-following example discussed later.

1.3.2.7. Power Supply and Power Distribution

The power system of the 3pi+ 32U4 robot is designed with multiple voltage regulation stages and protection circuits to provide stable power to each functional block. The main power source is a battery holder containing four AAA batteries. It can use NiMH batteries with a nominal voltage of 4.8V or alkaline batteries with a nominal voltage of 6V, providing the raw battery voltage VBAT.

The VBAT supply first passes through a reverse-polarity protection circuit and an electronic power switch to generate VSW. This voltage is then distributed to several regulators to produce three main voltage levels. The 8V (VM) supply is dedicated to the two DRV8838 motor drivers. The 5V supply powers the ATmega32U4 MCU, the logic sections of the motor drivers, the line sensors, collision sensors, and encoders. Meanwhile, the 3.3V supply is generated from the 5V

rail through an LDO and is used to power the IMU and the associated signal level-shifting circuits.

In addition, the system integrates a TPS2113A power multiplexer to automatically select between the 5V supply from the battery and the 5V supply from USB. This allows the robot to be connected to a computer for programming or testing the logic and sensor systems without relying on the battery. At the same time, the motors remain inactive when the main power switch is turned off.

1.3.3. Control system

The control system of the Pololu 3pi+ 32U4 processes sensor information and generates commands for the two drive motors. It can be described through the overall control architecture and the line-following control algorithm.

1.3.3.1. Control architecture

The Pololu 3pi+ 32U4 can be considered a centralized control system. The central controller receives signals from the sensors and encoders, processes this information, and sends control commands to the left and right motor drivers. The motor drivers then control the corresponding drive motors. The encoders can provide wheel-speed feedback to the controller.

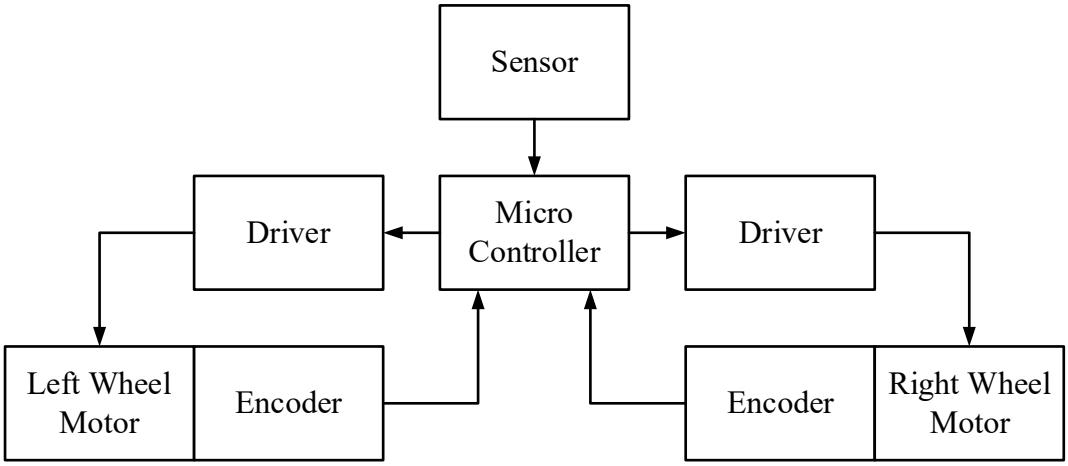


Figure 1.10. Centralized control architecture of the Pololu 3pi+ 32U4

The robot also includes quadrature encoders, an IMU, and bump sensors. These devices are available for motion feedback and additional sensing functions; however, they are not all used in the official line-following example.

1.3.3.2. Line-following control algorithm

The robot uses five line sensors to determine the position of the line relative to the center of the sensor array. The measured line position is compared with the desired position to obtain the tracking error e. A PD controller processes this error and generates a speed correction Δv[4].

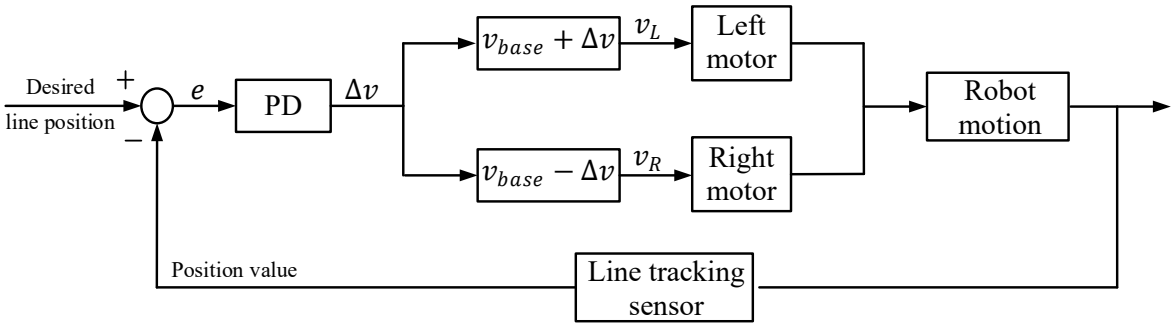


Figure 1.11. Line-following control algorithm of the Pololu 3pi+ 32U4

By increasing the speed of one wheel and decreasing the speed of the other, the robot changes its direction and returns toward the line. The line sensors continuously update the measured line position to form the feedback loop.

The motor encoders are not used for feedback control, even though both motors are equipped with encoders for measuring their actual rotational speed. The wheel speeds in this implementation are controlled directly by applying PWM commands to the motor drivers. Therefore, there is no additional closed-loop speed-control or PID loop for individually regulating the speed of each motor.

The IMU and bump sensors are also not used. Although the robot is equipped with an IMU containing an accelerometer, gyroscope, and magnetometer, as well as two front bump sensors, the current source code does not include any data acquisition or processing functions for these sensors.

1.4. Apollon 2025 line following robot

The robot was designed and manufactured by Akihiro Fujisawa, took part in the RoboRacing Competion Robotrace MM2025 and get the first prize[5].

1.4.1. Mechanical system

1.4.1.1. Mechanical principles

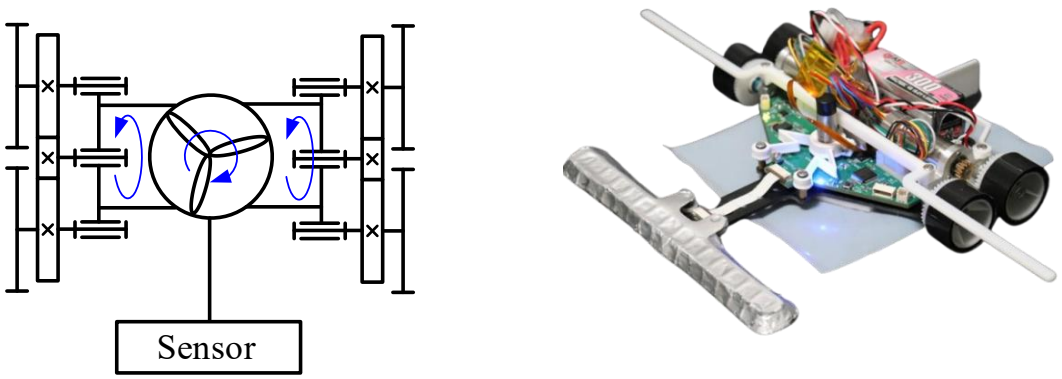


Figure 1.12. Apollon 2025 line following robot

The Apollon 2025 line-following robot is designed with a four-wheel drive mechanism, powered by two motors that transmit torque to all four wheels via a gear drive. The robot

employs a differential drive mechanism, steering through velocity differences between the two sides. It is equipped with four silicone wheels, providing high traction. In addition, the robot is equipped with a centrally mounted suction fan, responsible for generating negative pressure to increase the downward force on the line map, thereby preventing loss of stability at high operating speeds. The robot is additionally designed with lateral stabilizing wings extending to both sides, enhancing stability during cornering and preventing rollover. IR sensors are arranged in a horizontal array at the front of the robot.

1.4.1.2. Summary of basic mechanical specification

Table 1.3. Summary of basic mechanical specification

Specifications	Type	Q.ty	Value
Running motor	BLDC motor Maxon ECXSpeed 13M	2	–
Suction fan	BLDC motor Maxon ECXSpeed 8M	1	–
Wheel	Silicone seat tire	4	∅22×12
Driving structure	Differential	–	–
IR sensor	ROHM SCM - 014TB	16	–
	Viewing angle		10-20°
	Distance between each sensor		5mm
	Distance to the line map		4mm
Encoder	Magnetic Maxon ENX EASY (incremental/absolute)	2	4096 steps
Speed	Straight-line speed	–	11m/s
	Acceleration	–	$35m/s^2$
	Turning speed	–	2m/s

1.4.2. Electrical system

1.4.2.1. Electrical system architecture

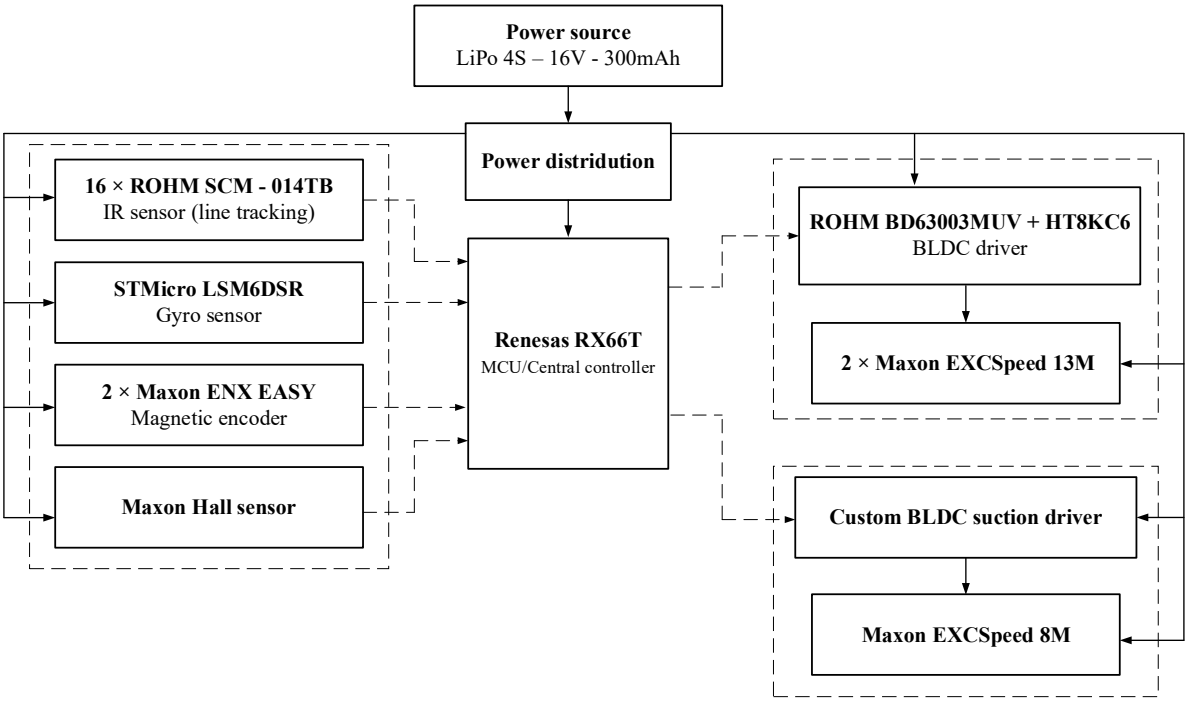


Figure 1.13. Electrical diagram

1.4.2.2. Microcontroller

The APOLLON robot uses the Renesas RX66T microcontroller as its central processing unit. According to the APOLLON 2025 technical document, the RX66T operates at a clock frequency of 160 MHz and is equipped with 512 KB ROM, 64 KB RAM, and 8 MB DataROM.

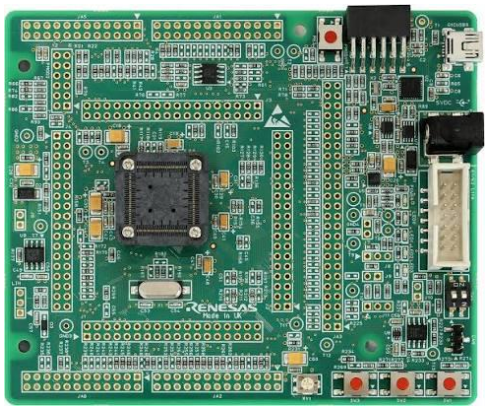


Figure 1.14. Renesas RX66T microcontroller

Main communication interfaces:

- ADC: The Analog-to-Digital Converter (ADC) is used to acquire the analog signals generated by the 16 ROHM SCM-014TB infrared line sensors. The reflected-light

intensity is converted into an electrical signal and then digitized for line-position processing.

- PWM: PWM signals are generated by the RX66T to control the motor driver. By adjusting the PWM command, the controller can regulate the motion of the left and right drive motors.
- GPIO: GPIO pins can be used for digital control and status signals, including enable signals and digital feedback signals from sensors such as Hall sensors or encoders, depending on the actual hardware configuration.
- I²C/SPI: The APOLLON robot uses an STMicroelectronics LSM6DSR IMU. In the current report, I²C or SPI is considered as the communication interface between the RX66T and the IMU.

1.4.2.3. Driver and actuator

The APOLLON robot employs a brushless motor drive system consisting of a motor driver, a power MOSFET stage, main drive motors, and a suction motor. The main components of the motor drive and actuator system are summarized in the following table.

The main drive system uses a ROHM BD63003MUV three-phase BLDC pre-driver together with external HT8KC6 N-channel MOSFETs. The driver receives logic and PWM control signals from the controller and controls the three-phase power stage that supplies the two Maxon ECXSpeed 13M drive motors. A separate custom BLDC driver is used for the ECXSpeed 8M suction motor.

Table 1.4. Motor driver function

Component	Model
Motor driver	ROHM BD63003MUV
Power MOSFET	ROHM HT8KC6
Main drive motors	2 × Maxon ECX Speed 13M
Suction motor	1 × Maxon ECX Speed 8M

Table 1.5. ROHM BD63003MUV specifications[5]

Parameter	Specification
Driver IC	ROHM BD63003MUV
Type	3-phase BLDC pre-driver
Supply voltage	$10.8 - 26.4\ V$
Logic input	$3.3\ /\ 5\ V$
Speed control	PWM
Commutation	120°

Rotor feedback	3 Hall sensor inputs
External power stage	N-channel MOSFET
Protection	Current limit, UVLO, OVLO, thermal protection

Table 1.6. ROHM HT8KC6 specifications[6]

Parameter	**Specification**
Device type	Dual N-channel Power MOSFET
Drain-source voltage V_{DSS}	$60\ V$
Drain current I_D	$15\ A$
On-resistance $R_{DS(on)}$	$\approx\ 23\ m\Omega$
Function	External power switching stage for the BLDC motor driver

Table 1.7. Maxon ECXSpeed 13M specifications[7]

Parameter	**Specification**
Motor type	$3-phase\ Brushless\ DC\ motor$
Diameter	$13\ mm$
Nominal voltage options	$18\ /\ 24\ /\ 36\ /\ 48\ V$
Nominal speed	$36{,}900-38{,}400\ rpm$
Nominal torque	$3.99-4.23\ mNm$
Nominal current	$0.419-1.20\ A$
Stall torque	$25.1-28.3\ mNm$
Stall current	$2.48-7.49\ A$
Maximum speed	$90{,}000\ rpm$

Table 1.8. Maxon ECXSpeed 8M specifications[8]

Parameter	**Specification**
Motor type	$3-phase\ Brushless\ DC\ motor$
Diameter	$8\ mm$
Nominal voltage options	$6\ /\ 12\ /\ 24\ V$
Nominal speed	$24{,}200\ -\ 32{,}300\ rpm$

Nominal torque	$0.877 - 0.917\ mNm$
Nominal current	$0.198 - 0.687\ A$
Stall torque	$3.04 - 3.92\ mNm$
Stall current	$0.755 - 1.98\ A$
Maximum speed	$50{,}000\ rpm$

1.4.2.4. Line sensor system

The APOLLON robot uses an array of 16 ROHM SCM-014TB infrared phototransistors for line detection. The SCM-014TB is a surface-mount, lens-type phototransistor with a peak sensitivity wavelength of approximately 800 nm and a compact package size of approximately $3.0 \times 1.5 \times 2.2$ mm, with an operating angle of approximately $10 - 20°$[9] .

The SCM-014TB provides an analog output in the form of a varying collector current corresponding to the received light intensity. Through a current-to-voltage conversion circuit, this current variation is converted into a voltage signal and then digitized by the ADC of the Renesas RX66T microcontroller. The RX66T processes the signals from the 16 sensors to estimate the relative position of the line, which is subsequently used as an input for the robot motion-control system.

1.4.2.5. Feedback and additional sensors

a. Encoder

The APOLLON robot is equipped with two Maxon magnetic encoders for drive-system feedback. The APOLLON documentation refers to the Maxon ENX EASY 10/16 mm series, which uses a magnetic sensing principle and can provide incremental or absolute position information.

In incremental mode, the ENX EASY provides three digital channels: A, B, and index I. Channels A and B generate square-wave pulses with a phase difference, allowing the controller to determine the rotation direction and speed of the motor shaft. The index channel provides one reference pulse per revolution. The incremental resolution can be configured from 1 to 1024 counts per turn (cpt).

The encoder signals are fed back to the controller to monitor the actual motion of the drive system and support position and speed control.

Table 1.9. Maxon ENX EASY specifications[10]

Parameter	Specification
Model	Maxon ENX EASY
Quantity	2
Action type	Magnetic

Signal type	Incremental / Absolute
Resolution	1 … 1024 cpt / 4096 steps

b. Hall Sensor

The APOLLON robot uses two Maxon Hall sensors to detect the magnetic field associated with the motor rotor. The Hall sensors provide information about the rotor position, which is used to support the commutation process of the brushless DC motors.

The sensors provide digital rotor-state signals to the motor-control system. However, the available APOLLON documentation does not specify the exact Hall sensor model or detailed electrical characteristics.

c. LSM6DSR

The APOLLON robot uses the STMicroelectronics LSM6DSR as its inertial measurement unit (IMU). This sensor is used to provide motion information that supports the monitoring of the robot's dynamic state during operation. The main specifications of the LSM6DSR are summarized in the following table.

Table 1.10. LSM6DSR specifications[11]

Parameter	Specification
Sensor model	STMicroelectronics LSM6DSR
Quantity	1
Sensor type	6-axis IMU
Accelerometer	3-axis
Gyroscope	3-axis
Measured data	Linear acceleration and angular velocity
Output type	Digital
Communication interface	I²C / SPI

1.4.2.6. Power supply system

The APOLLON robot uses a GAONENG 4-cell (4S) LiPo battery as the main power source for the entire system. The main battery specifications are summarized in the following table.

Table 1.11. Power suppy specifications[12]

Parameter	Specification
Battery type	LiPo
Manufacturer	GAONENG
Cell configuration	4S

Capacity	300 mAh
Main system voltage	16 V
Function	Main power supply for the robot

1.4.3. Control system

The APOLLON robot uses a centralized control architecture, in which the Renesas RX66T acts as the main controller. Signals from the line sensor array and motion-feedback sensors are collected and processed by the microcontroller. Based on the detected line position and the current motion state, the RX66T calculates the required commands for the left and right drive motors.

1.4.3.1. Control architecture

The control commands are then transmitted to the motor driver and power stage to adjust the speed of the two drive sides. The difference between the left and right wheel speeds allows the robot to change direction and follow the track. The two encoders provide motion feedback to the controller, allowing the drive state to be continuously updated during operation.

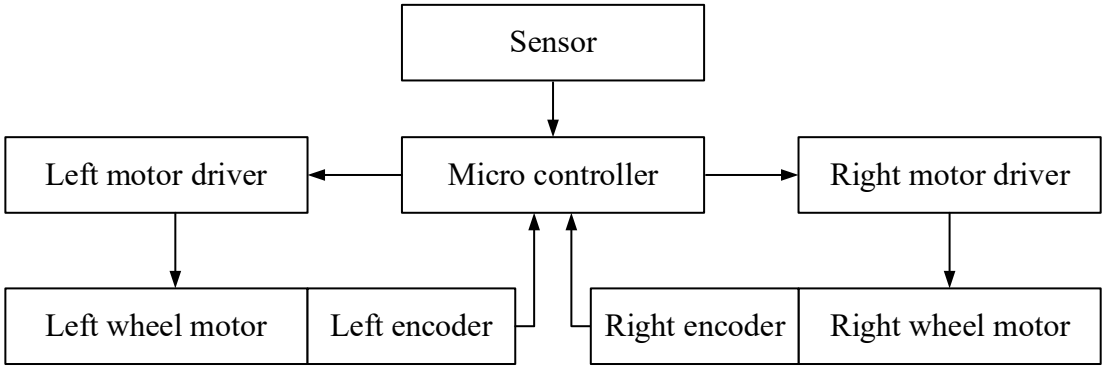


Figure 1.15. Structure of centralized control

1.4.3.2. Control Algorithm

a. Line Tracking Algorithm

The line-tracking control process begins by acquiring data from the line sensors and IMU to determine the position of the track relative to the robot. When the line is detected, the controller determines the required difference between the left and right wheel speeds and sends the corresponding commands to the motor drivers. Encoder feedback is used to monitor the robot motion, and the control cycle is continuously repeated until the course is completed.

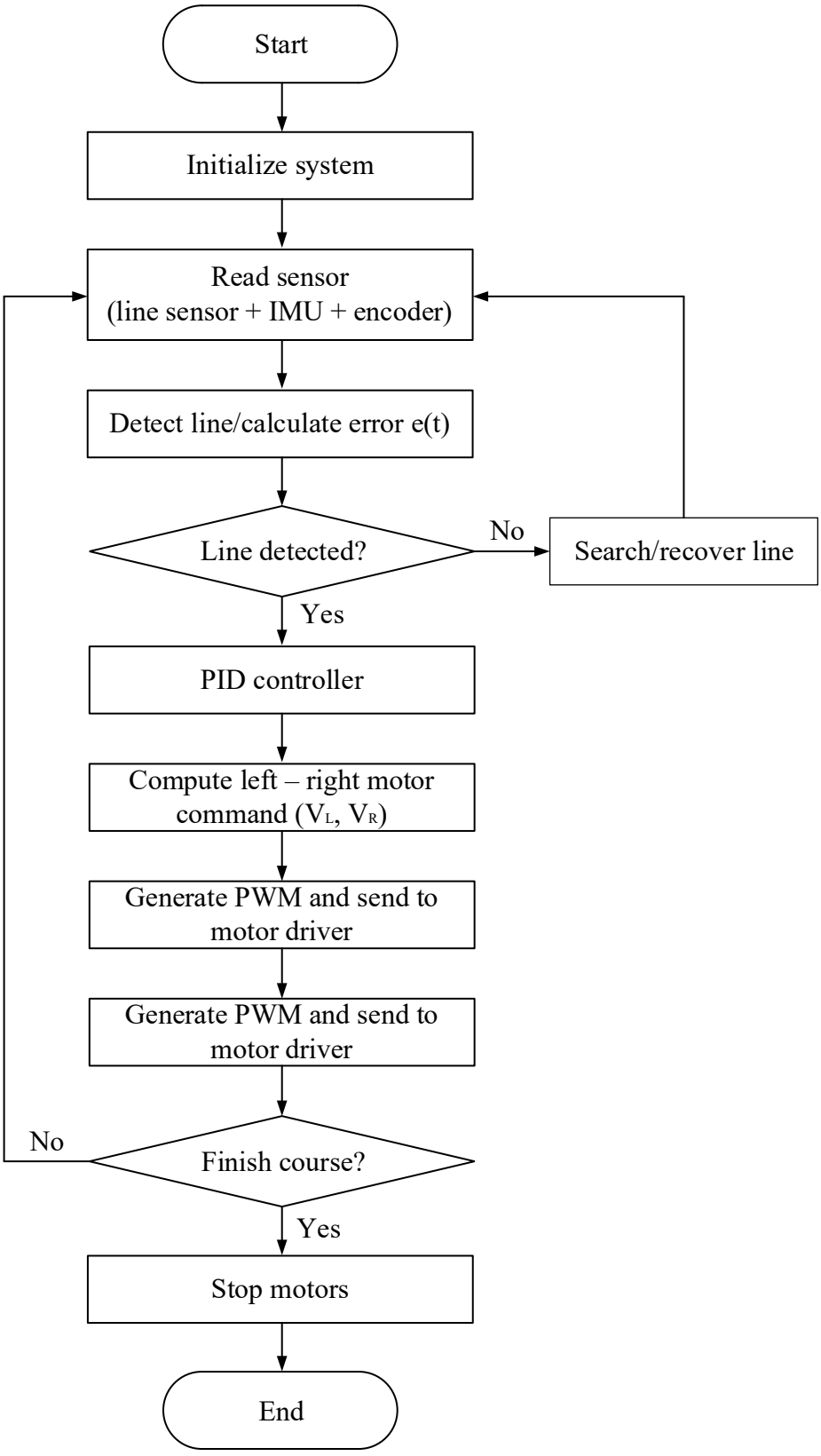


Figure 1.16. Flowchart of APOLLON line-following control algorithm

b. Motor Control Algorithm

In the PID-based control process, the controller first determines the line-position error $e(t)$ from the sensor data. This error is processed by the PID controller to generate the required correction signal. The controller then calculates the left and right motor commands (v_L, v_R) and generates the corresponding PWM signals for the motor drivers. This process is repeated continuously to adjust the robot direction and maintain line tracking.

1.5. Problem Definition

1.5.1. Main functions

The control system must maintain the line-following error within the allowable limit while tracking a black line on a white background. The reference point is defined as the midpoint of the robot's front edge.

Starting from a predefined position, it must autonomously move along the designated path and stop at the target position, represented by one of the intersections, while maintaining the stopping error within the allowable limit.

The robot must also be able to detect a cylindrical obstacle placed at an equal distance from an intersection on either of the two possible paths. Once the obstacle is detected, the control algorithm must automatically redirect the robot to the alternative path, allowing it to safely bypass the obstacle and continue its operation.

1.5.2. System specifications

Table 1.12. System specifications

Feature	Symbol	Value
Line width	w_l	26mm
Minimum turning radius	R_{min}	500mm
Minimum speed	v_{min}	0.1m/s
Maximum design speed	v_{max}	2m/s
Allowable line-following error	e_{max}	$\pm$3mm
Allowable stopping error	e_{stop}	$\pm$5mm
Obstacle diameter	d_{obs}	50mm
Selected obstacle height	h_{obs}	200mm

1.6. Project objectives

The overall objective of this project is to research, design, simulate, and fabricate a complete prototype of an autonomous line-following robot. The system is required to accurately follow a predefined path while being capable of detecting obstacles and autonomously making decisions to change direction when an obstacle is encountered at an intersection. The project aims to deliver the following hardware, software, and design documentation components:

Mechanical design: Complete the 2D CAD assembly drawings for the entire mechanical system. The design includes the chassis, motor mounting brackets, drive and steering mechanisms, and mounting clamps for the sensor array, with an emphasis on optimizing the overall space utilization and weight.

Electrical and electronic system: Successfully design and fabricate dedicated printed circuit boards (PCBs). The electronic system must integrate all major functional blocks, including the voltage boost/buck power supply, central microcontroller, motor power control, and sensor-array interface.

Control algorithms and software: Develop a mathematical model of the robot's planar kinematics. Based on this model, develop the velocity controller and trajectory-tracking controller. The controllers must be validated through simulation to optimize their parameters before being compiled into source code and deployed to the microcontroller of the physical system.

1.7. Scope of the project

This project focuses on addressing the core problem of an autonomous guided vehicle (AGV) system based on a line-following algorithm. To optimize research resources and ensure the feasibility of the implementation, the operating capabilities of the system and the problems beyond the scope of the study are clearly defined.

Model Limitations

Closed operating environment: The robot is designed to operate reliably only in an indoor environment on a flat and dry surface. Its performance may become unstable or inaccurate when operating on uneven surfaces, surfaces with low traction, or outdoors. The robot follows a predefined and relatively simple path layout and does not have the capability to determine the desired destination when multiple branches are present.

Sensitivity to environmental interference: The line-following sensors can be affected by high-intensity ambient light or shadows. The obstacle-detection sensor may also produce inaccurate measurements when the obstacle has an inclined surface that causes the transmitted signal to be reflected away from the sensor, or when the obstacle surface absorbs infrared or ultrasonic signals.

Limited to simple static obstacles: The current control program is designed to handle only stationary obstacles placed within predefined branching areas, with limited and predetermined positions and distances. The robot does not yet support dynamic obstacles crossing its path or automatically avoiding unexpected obstacles that appear along straight sections of the route.

1.8. Organization of the report

The report is organized according to the design process of a mechatronics system, from requirement surveying and alternative selection to design, simulation, and experimentation:

- Chapter 1. Overview: Presents the concept of a line-following robot and surveys applied locomotion mechanisms, electrical systems, and control methods. From there, it defines the objectives, technical requirements, design parameters, and scope of the project.
- Chapter 2. Analysis and Design Selection: Proposes alternatives for the mechanical structure, electrical/electronic system, and control architecture; analyzes their pros and cons; and selects the appropriate solution for the robot.
- Chapter 3. Mechanical Design: Determines the geometric and dynamic parameters of the robot, calculates the transmission system requirements, selects the motors, and designs the mechanical components of the robot.
- Chapter 4. Electrical System Design: Develops the electrical system architecture, selects and designs the sensor circuits, microcontroller, power circuits, power supply system, and the connections between functional blocks.
- Chapter 5. Controller Design and Modeling: Constructs the kinematic model of the robot, determines the line-tracking error variables, and designs the motion controller as well as the motor velocity controller.
- Chapter 6. Control Algorithm: Presents the flowcharts and overall control algorithms, including sensor reading, line tracking, obstacle detection, intersection recognition, and appropriate branch selection.
- Chapter 7. Simulation: Simulates the motion and the controller to evaluate trajectory tracking capability, tracking errors, and system response prior to conducting physical experiments.
- Chapter 8. Experiment and Results: Presents the completed robot prototype, experimental conditions, and procedures; collects experimental data; and evaluates the results against the predefined design requirements.
- Chapter 9. Conclusion: Summarizes the achieved results, evaluates the degree of fulfillment of the initial objectives, points out limitations, and proposes directions for future development.

CHAPTER 2. DESIGN PROPOSAL

2.1. Mechanical proposal

Full-filling the project's objectives, specifications and main function, we proposes the below configurations.

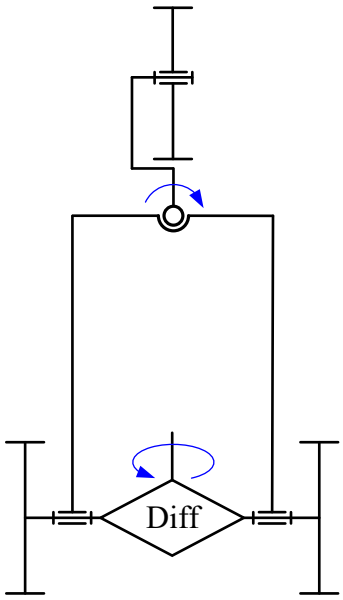


Figure 2.1. Proposal mechanisms

This configuration is a hybrid variant combining differential mechanism and tricycle mechanism. It leverages the flexible steering capability of the tricycle mechanism together with the differential speed between the two wheels resulting in highly agile mobility.

This proposed configuration eliminates the use of two rear motors as in traditional differential mechanism, which would otherwise introduce multiple fault and increase overall mass, making it unsuitable for the design specifications of a high-speed robot.

2.2. Electrical proposal

The proposed electrical system consists of a microcontroller (MCU), a line-tracking sensor, an obstacle detection sensor, a steering motor, a rear-wheel driving motor, an encoder, and a power supply system.

2.2.1. Controller platform proposal

A microcontroller-based control system is proposed due to the robot's compact size, limited onboard power, and real-time control requirements.

Compared with PLCs and general-purpose computers, a microcontroller provides a compact, low-power solution with integrated peripherals for sensor acquisition, encoder feedback, and motor control. Its processing capability is sufficient for the proposed sensor-based control architecture.

A single MCU is used to process sensor data, execute control algorithms, and generate commands for the steering and driving motors. The specific MCU will be selected in Chapter 4 based on the required processing capability and peripheral interfaces.

2.2.2. Line tracking sensor proposal

An infrared reflectance sensor array is proposed to detect the black line on the white background and estimate its position relative to the robot.

The sensor arrangement must accommodate the specified line width of 26 mm and provide sufficient measurement accuracy for the allowable line-tracking error of ± 3 mm.

The number of sensors, spacing, mounting height, and measurement error will be determined through analysis, calibration, and experimentation in Chapter 4.

2.2.3. Obstacle detection sensor proposal

A non-contact distance sensor is proposed to detect obstacles in the designated areas.

The sensor must detect the specified cylindrical obstacle with a diameter of 50 mm and a height of 100–200 mm. Its detection range, field of view, and response time must allow the robot to identify the blocked path before making a navigation decision.

The sensor specifications, mounting position, and measurement accuracy will be determined in Chapter 4.

2.2.4. Actuator and feedback configurations

The proposed system uses two actuators with different functions.

The steering motor controls the steering angle to adjust the robot's direction, while the rear-wheel motor provides the driving force. The MCU sends steering commands directly to the steering actuator and controls the rear-wheel motor through a motor driver.

An encoder mounted on the rear-wheel motor provides rotational feedback for closed-loop speed control.

The required motor specifications will be determined from the mechanical calculations. The motor driver and encoder specifications will subsequently be determined based on the selected motor and control requirements.

2.2.5. Power supply

A battery-powered system is proposed with two power distribution branches.

The control voltage regulator supplies the MCU, sensors, and encoder. The dynamic voltage regulator supplies the motor driver and steering actuator.

This arrangement provides regulated power for the control electronics and separates their supply distribution from the actuator power branch.

The battery specifications, voltage regulators, filtering, and protection circuits will be determined in Chapter 4.

2.2.6. Electrical system architecture

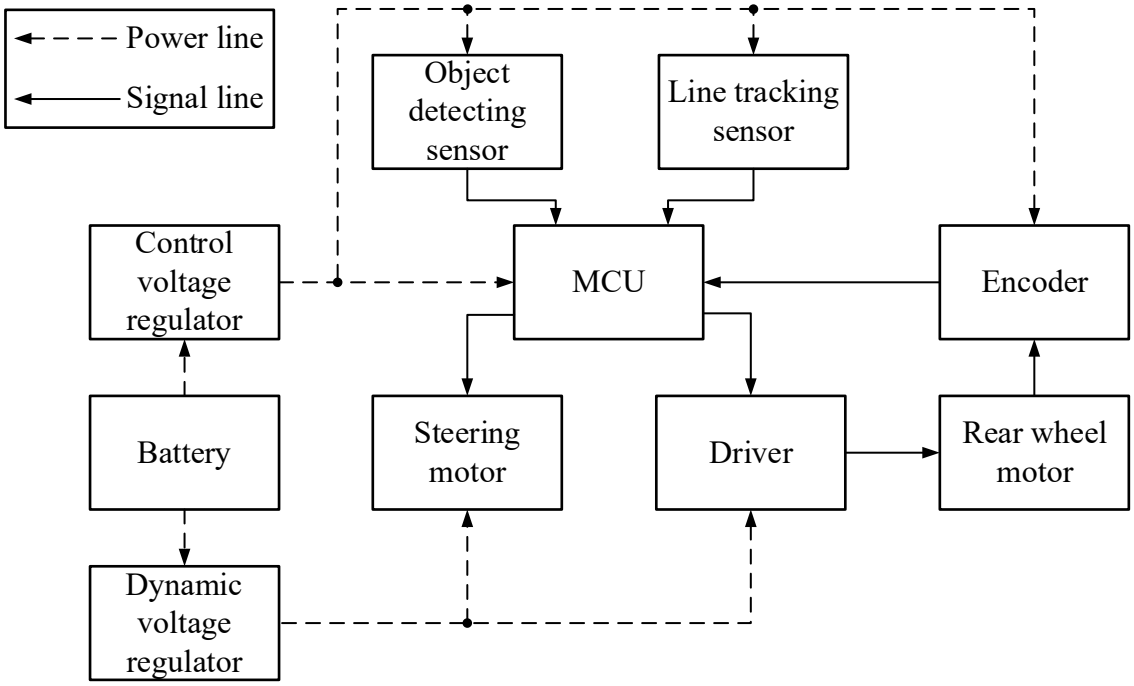


Figure 2.2. Electrical system diagram

2.3. Control proposal

2.3.1. Overall Control Architecture

The proposed design adopts a Centralized Control architecture. In this architecture, the entire control cycle—including data acquisition, algorithm processing, and control command generation—is executed by a single microcontroller.

The system data flow is organized as follows:

- Data Acquisition Unit: The central microcontroller directly acquires input signals from three main sources: the line-tracking sensor array; the distance sensor; and the encoder pulse signals.
- Central Processing Unit: The acquired sensor data are processed by the control algorithms programmed inside the microcontroller. These algorithms include the trajectory-tracking control law and the motor-speed control law.
- Signal Output Unit: Based on the calculated control commands, the microcontroller directly generates Pulse Width Modulation (PWM) signals for the motor driver to regulate the power delivered to the drive motors. At the same time, directional control signals are generated for the steering mechanism, thereby completing the closed-loop operation of the robot.

2.3.2. Control Methods

The system employs a Cascade Control architecture consisting of two controllers operating simultaneously to address the trajectory-tracking and motor-speed control requirements.

Trajectory Tracking Controller (Outer Loop): Nonlinear control based on Lyapunov stability theory.

- Input Variables: The lateral error (e_2) and heading error (e_3), which are derived from the signals acquired from the line-tracking sensor array.
- Output Variables: Reference linear velocity (v_c) and reference angular velocity/steering angle (ω_c/γ_c). These signals serve as the setpoints for the inner control loop.

Motor Speed Controller (Inner Loop): Proportional–Integral–Derivative (PID) Control.

- Input Variable: The velocity error (e_v), calculated as the difference between the reference velocity generated by the outer-loop controller and the actual wheel rotational velocity measured by the encoders.
- Output Variable: The PWM duty cycle. This value is directly converted into a PWM signal and supplied to the power driver, which regulates the effective voltage applied to the motor. Consequently, the rotational speed of the motor is controlled to follow the velocity reference generated by the outer loop.

2.3.3. Control System Block Diagram

General block diagram of the control system:

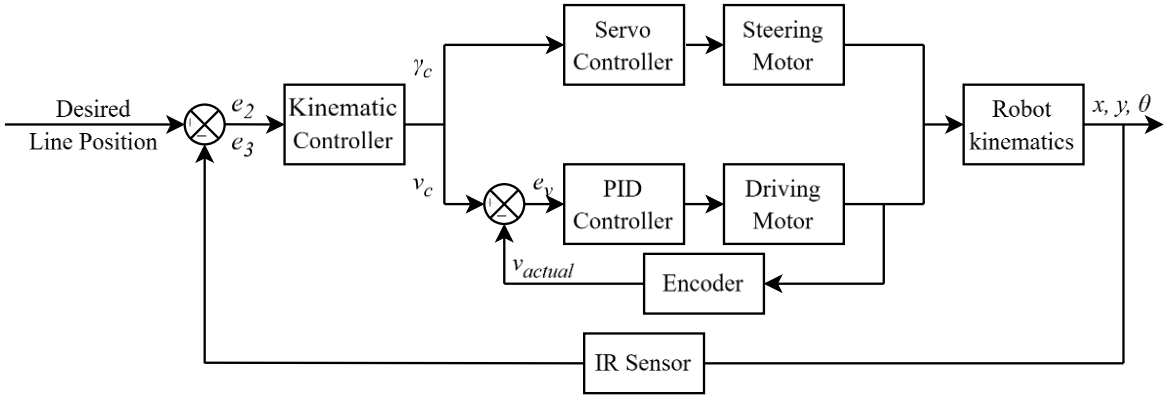


Figure 2.3. Control system block diagram

Description of the components in the block diagram:

The system employs a cascade control architecture, divided into an outer kinematic loop for trajectory tracking and two independent inner loops for actuator control:

- Trajectory Tracking (Outer Loop): The Lyapunov Controller processes the lateral error (e_2) and heading error (e_3) derived from the IR sensor array. It calculates the reference steering angle (γ_c) and reference linear velocity (v_c) required to guide the robot along the path.
- Steering Control (Inner Loop 1): The reference angle (γ_c) is transmitted to the Servo Controller, which directly drives the Steering Motor to achieve the desired wheel angle.
- Velocity Control (Inner Loop 2): The velocity error (e_v)—the difference between the reference velocity (v_c) and the actual encoder feedback (v_{actual})—is processed by a PID Controller to regulate the Driving Motor.

- Robot Kinematics & Feedback: The mechanical outputs of both motors govern the robot's physical pose (x, y, θ). The IR sensors continuously measure the new relative positional errors, closing the outer feedback loop.

2.3.4. Algorithm

The flowchart illustrates the overall operating algorithm of the robot. After initialization, the robot waits for a start signal from the system. Once the signal is received, the robot performs obstacle detection and begins the line-tracking process. During movement, the system continuously monitors the line status and determines whether the robot has reached an intersection. If no intersection is detected, the robot continues following the line. When an intersection is detected, the robot executes the predefined intersection-handling algorithm and then resumes line tracking. This process continues until the line can no longer be detected, at which point the robot stops and the operating process is completed.

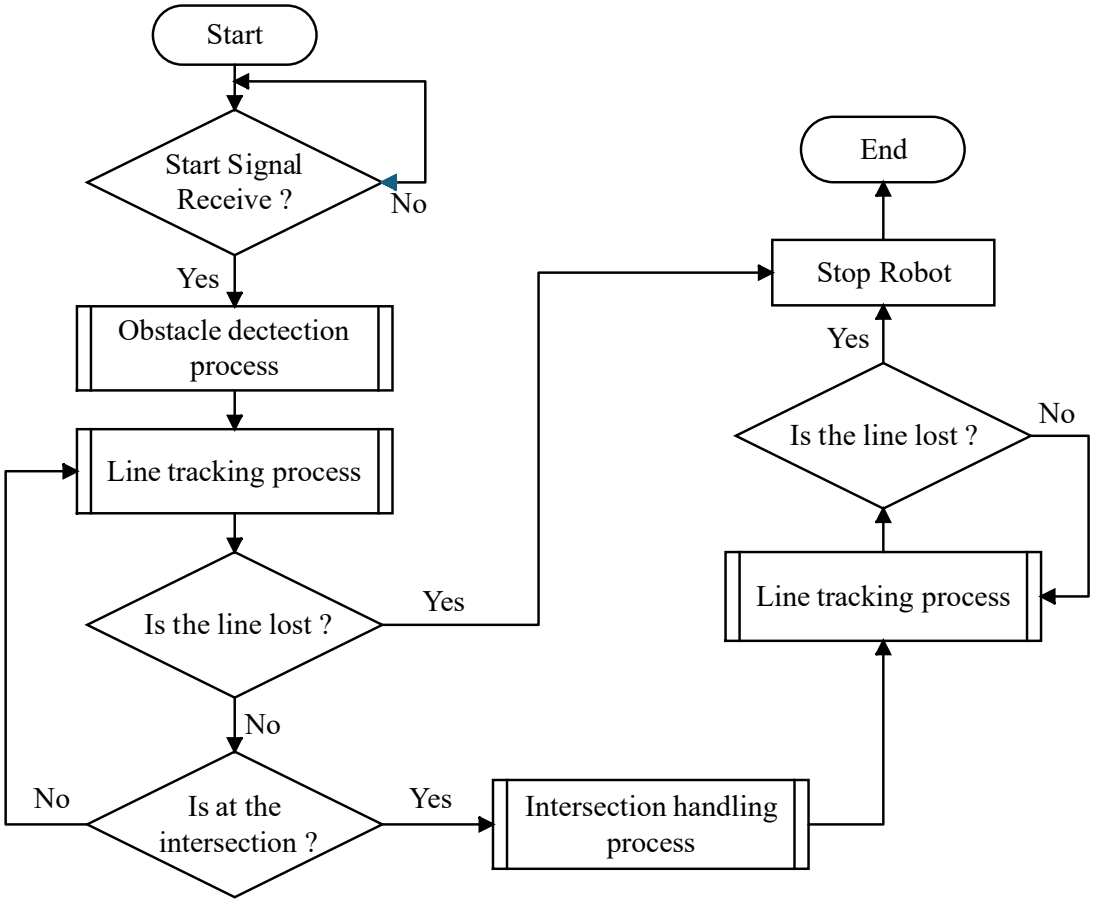


Figure 2.4. System operational control flowchart

CHAPTER 3. MECHANICAL DESIGN

3.1. Input parameters

From the overall assessment, the chosen design is a three-wheel mobile robot with rear drive wheels utilizing WLToys 144001 differential gearbox, connect to driving DC motor through a flexible coupling. Since the robot steering control through a stepper motor actively provides stability. The zero trail configuration is selected for the front steering geometry (steering axis pass through tire). Letting stepper fully control steering angle without fighting with restoring force and simplify kinematics.

Table 3.1. Input parameters

Parameters	Value	Unit
Maximum velocity, v	2	m/s
Robot's mass, M	1.5	kg
Friction coefficient, μ	0.4	-
Safety factor h_s	1.5	-
Cardan efficiency, η_c	0.96	-
Gear efficiency, η_g	0.95	-
Differential gearbox ratio, u_{dg}	2.5	-

3.2. Structural constraints

From the overall assessment, the chosen design is a three-wheel mobile robot with rear drive wheels power by a differential gearbox and a front wheel for steering. This kinematic model as below.

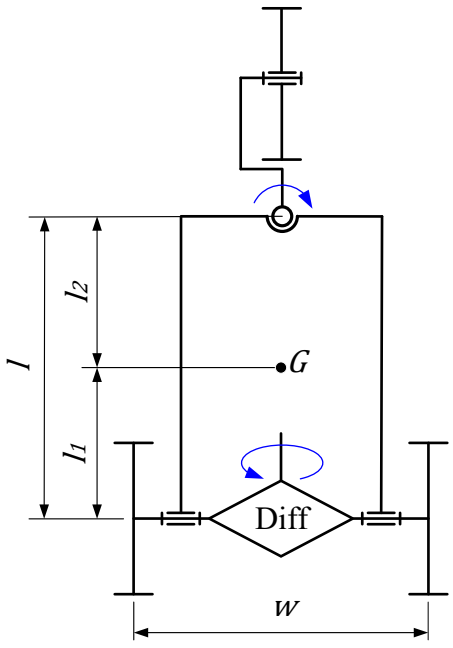


Figure 3.1. Kinematics diagram of the mechanical model

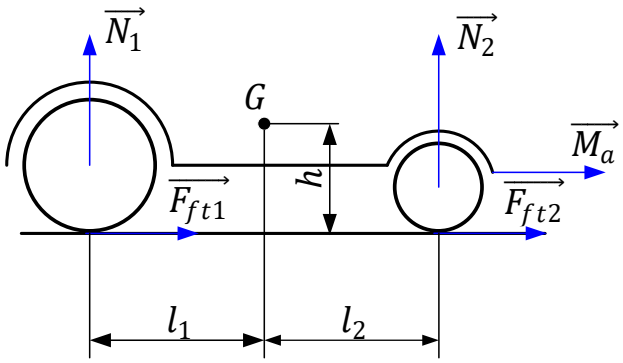


Figure 3.2. Dynamic analysis

Based on the Newton II Law and equilibrium equations, we have a system of dynamic equations of the vehicle are as follow:

$$\begin{cases} \sum \overrightarrow{F_x} = M\vec{a} \\ \sum \overrightarrow{F_y} = 0 \\ \sum \overrightarrow{F_z} = 0 \\ \sum M_{/G} = 0 \end{cases} \Leftrightarrow \begin{cases} F_{ft1} + F_{ft2} = Ma \\ N_1 + N_2 = Mg \\ N_1 l_1 - N_2 l_2 - F_{ft1} h - F_{ft2} h = 0 \end{cases} \quad (3.1)$$

Where: G: center of mass of the whole system.

$\overrightarrow{F_{ft1}}, \overrightarrow{F_{ft2}}$: friction force acting on rear and front wheels.

$\overrightarrow{N_1}, \overrightarrow{N_2}$: normal force acting on rear and front wheels.

l_1, l_2: distance from G to rear and front wheel center axes.

h: distance from G to the ground.

From equation (3.1), we derive the acceleration as a function of parameters:

$$(mg - N_2)l_1 - N_2 l_2 - Mah = 0 \Leftrightarrow a = \frac{gl_1}{h} - \frac{N_2(l_1 + l_2)}{mh} \quad (3.2)$$

From this equation, we can get the correlation between and the vehicle's dimension:

- The greater the mass M, the more it affects to the acceleration (Newton II Law).
- The smaller h, the larger a will be, so smaller wheels or lower ground clearance are more favorable for acceleration.
- The greater distance from the rear wheel axis to the center of gravity G (l_1), the greater the acceleration a.

3.3. Preliminary robot dimensions

Preliminary, we choose the ratio between width w and the length l is 1.5. With the preliminary width $w = 150\text{mm}$, we get the length:

$$l = 1.5 \times w = 1.5 \times 150 = 225\text{mm}$$

3.4. Slippery condition

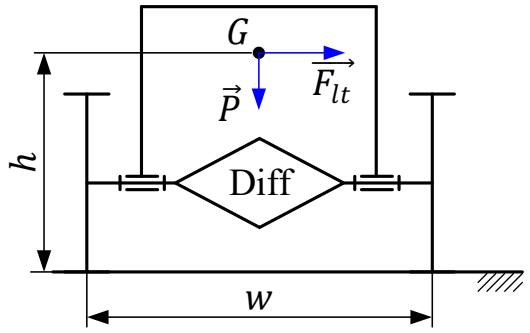


Figure 3.3. Slippery analysis

To avoid flipping over, the moment generated by the gravity about the center G must be greater than the moment generated by the centrifugal force about G. The condition at the wheel contact point is:

$$P\frac{w}{2} \geq F_{lt}h \Leftrightarrow Mg\frac{w}{2} \geq \frac{Mv^2}{R}h \Leftrightarrow h \leq \frac{wgR}{2v^2} = \frac{0.15 \times 9.8 \times 0.5}{2 \times 2^2} = 0.092\text{m} = 92\text{ mm} \quad (3.3)$$

Where: w: Robot's rear width, $w = 0.15$ m.

R: Max curve radius, $R = 0.5$ m.

v: Robot's velocity, $v = 2$ m/s.

Preliminary select $h = 50\text{mm} = 0.05\text{m}$, re-checking the maximum velocity that the robot can achieve when turning:

$$v \leq \sqrt{\frac{wgR}{2h}} \Rightarrow 2m/s < \sqrt{\frac{0.15 \times 9.8 \times 0.5}{2 \times 0.05}} = 2.711\ m/s \quad (3.4)$$

3.5. Vertical flip condition

During acceleration, the relationship between acceleration and vehicle dimensions must ensure that the front wheel keeps contact the ground, this condition affects if the line tracking sensor work well or not. Therefore, the condition $N_2>0$ must be satisfied.

From the dynamic equation (3.1):

$$\begin{cases} N_1 l_1 - N_2 l_2 - Mah = 0 \\ N_1 + N_2 = Mg \end{cases} \Rightarrow \begin{cases} N_1 = M\dfrac{gl_2 + ah}{l_1 + l_2} \\ N_2 = M\dfrac{gl_1 - ah}{l_1 + l_2} \end{cases} \quad (3.5)$$

Now apply the condition to N_2:

$$N_2 > 0 \Leftrightarrow M\frac{gl_1 - ah}{l_1 + l_2} > 0$$

$$\Rightarrow gl_1 - ah > 0 \Leftrightarrow a < \frac{gl_1}{h} = \frac{9.8 \times 0.05}{0.05} = 9.8\ m/s^2$$

Where $l_1 = 0.05\text{m} = 50\text{mm}$ is the distance from the rear wheel axis to the center of mass.

Select the acceleration $a = 2.5\text{m/s}^2$.

3.6. Driving wheel selection

The rotation speed of the wheel is affected by the wheel size:

$$n = \frac{60v}{2\pi R_w} = \frac{60 \times 2}{2\pi R_w} = \frac{60}{\pi R_w} \tag{3.6}$$

The required motor speed:

$$n_{motor} = nu = \frac{60}{\pi R_w} u = \frac{60}{\pi R_w} \times 1.5 = \frac{90}{\pi R_w} \tag{3.7}$$

Where, R_w: wheel diameter (m)

As the transmission ratio is fixed, the wheel size will affect the rotation speed of motor. If the necessary motor speed is too high, torque can not satisfy; and if the motor speed is too low, it will increase the wheel size

Set a study table about the relationship between the motor speed and wheel size. Select 4 common wheel size for this table

Table 3.2. Relationship between motor speed and wheel size

Wheel size (m)	Motor speed (RPM)
0.025	≈ 1910
0.0325	≈ 1470
0.04	≈ 1194
0.065	≈ 735

Based on the table, select wheel with radius $0.04m$ or diameter $\emptyset 80mm$ for the balance between motor speed and wheel size.

3.7. Driving motor selection

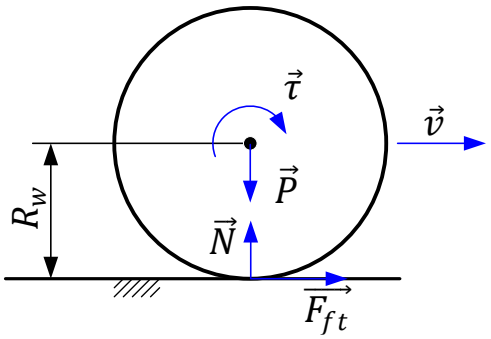


Figure 3.4. Rear wheel in motion analysis

Considering the wheel as solid circular disk, the moment of inertia:

$$I = \frac{1}{2} m_w R_w^2 \tag{3.8}$$

The inertia moment equilibrium equation:

$$\tau - F_{ft} R_w = \alpha I \Leftrightarrow \tau = \alpha I + F_{ft} R_w \tag{3.9}$$

Where $\alpha = a/R_w$ is the angular acceleration and R_w is the wheel radius

The no-slip condition when the motor rotates gives:

$$\tau \leq \alpha I + F_{ft} R_w$$

$$\tau \leq \frac{1}{2} a m_w R_w + \mu g R_w \left(\frac{1}{2} M + m_w\right)$$

$$\Leftrightarrow \tau \leq \frac{1}{2} \times 0.05 \times 1 \times 0.04 + 0.4 \times 9.8 \times 0.04 \times \left(\frac{1}{2} \times 1.5 + 0.05\right)$$

$$\tau \leq 0.126 \text{ (Nm)} \approx 1.284 \text{ (kg.cm)}$$

The required motor torque for no-slip condition (two wheels):

$$\tau_{motor\ no-slip} \leq \frac{2\tau}{u} = \frac{2 \times 0.126}{2.5} = 0.101\ Nm \approx 1.030 \text{ (kg.cm)}$$

From the horizontal force equilibrium equation:

$$F_{ft} = \frac{(2m_w + M)a}{2} \tag{3.9}$$

Substituting (3.9) into (3.7):

$$\tau = \frac{1}{2} a m_w R_w + \frac{(2m_w + M) a R_w}{2} = a R_w \frac{3m_w + M}{2}$$

The torque on the motor shaft:

$$\tau_{motor} = \frac{2\tau}{u} = \frac{2 a R_w}{u} \times \frac{3m_w + M}{2}$$

$$\Leftrightarrow \tau_{motor} = \frac{2 \times 1 \times 0.04}{2.5} \times \frac{3 \times 0.05 + 1.5}{2} \times 1.5$$

$$\Leftrightarrow \tau_{motor} = 0.040 \text{ (Nm)} \approx 0.408 \text{ (kg.cm)} < 1.030 \text{ (kg.cm)}$$

The required motor speed:

$$n_{motor} = \frac{60v}{2\pi R_w} u = \frac{60 \times 2}{2\pi \times 0.04} \times 2.5 \approx 1194 \text{ (RPM)}$$

The required motor power, included safety factor, efficiency of differential gearbox, outer gear drive and cardan shaft:

$$P = \frac{\tau_{motor} n_{motor}}{9.55 \times \eta_g^2 \times \eta_c} = \frac{0.040 \times 1194}{9.55 \times 0.95 \times 0.96} \approx 5.48 \text{ (W)}$$

The motor power above is the necessary mechanical motor power. The power mentions in the motor data-sheet is the electrical power. So the mechanical power of selected motor need to be calculated

Assume that the rated load speed and torque are the value after efficiency loss:

$$P_{selected\ motor} = \frac{\tau_{selected} n_{selected}}{9.55} = \frac{\tau_{selected} n_{selected}}{9.55}$$

Table 3.3. Required parameters for rear wheel motor

Parameters	**Value**	**Unit**
Maximum rotation speed	1194	RPM
Required torque	0.408	kg. cm
Power	5.48	W

Based on these criteria, we select the DC motor JGB37-3530-24100 with the following specifications:

Table 3.4. JGB37-3530-24100 DC motor parameters

Parameters	**Value**	**Unit**
No-load speed	1600	RPM
Rated speed under load	1231	RPM
Rated torque	0.5	kg. cm
Rated voltage	6.32	V
Rated current	24	A
Rated power	0.63	W

3.8. Steering motor selection

Assume that the robot is turning along a curve of radius R.

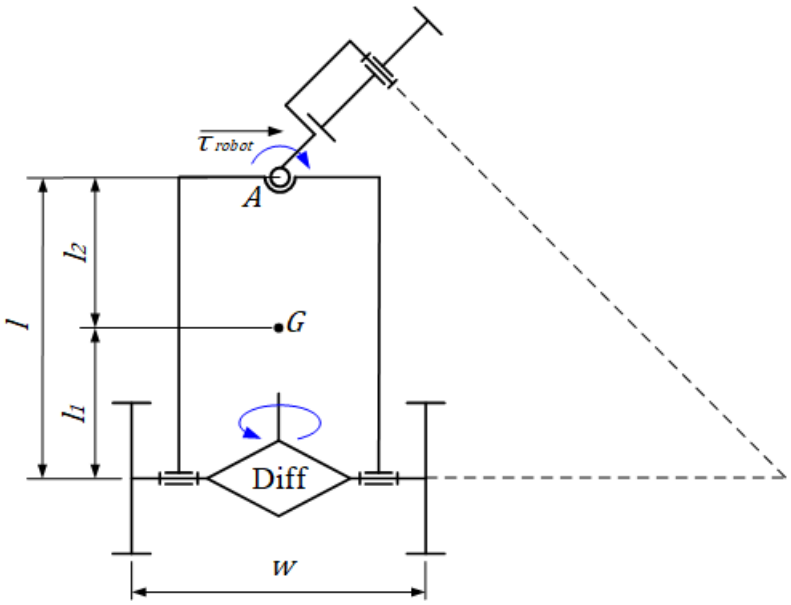


Figure 3.5. Steering analysis

Table 3.5. Steering parameters

Parameters	Value	Unit
Turning radius, R	500	mm
Distance from rear to steering wheel, l	225	mm
Rear width, w	150	mm

The angular acceleration of the model:

$$\alpha = \frac{a}{R} = \frac{1.5}{0.5} = 3\ rad/s^2$$

The moment of inertia about steering point A:

$$I = I_{body} + I_{rear} + I_{steering} = \frac{Ml^2}{12} + \frac{2m_w w^2}{12} + \frac{m_w R_w^2}{4} \quad (3.11)$$

The necessary torque for the robot steering at point A:

$$\tau_{steering} = \alpha I = \alpha\left(\frac{Ml^2}{12} + \frac{2m_w w^2}{12} + \frac{m_w R_w^2}{4}\right)$$

$$= 3 \times \left(\frac{1 \times 0.225^2}{12} + \frac{2 \times 0.03 \times 0.15^2}{12} + \frac{0.03 \times 0.04^2}{4}\right) = 0.013\ Nm = 0.133\ kg.cm$$

Since the steering mechanism is a 4 bar linkage with link 2,4 and link 1,3 each equals each other, leading to a parallelogram, assuming the efficiency is 100% resulting:

$$\omega_1 = \omega_3 \Leftrightarrow \tau_{servo} = \tau_{steering} = 0.133 kg.cm$$

Based on the required torque, we select the RC servo MG90S with a rated torque of 2kg.cm

CHAPTER 4. ELECTRICAL SYSTEM DESIGN

4.1. Overall electrical system

Based on the electrical architecture proposed in Chapter 2, the system consists of a microcontroller (MCU), a line-tracking sensor array, an obstacle detection sensor, a steering actuator, a rear-wheel driving motor, an encoder, a motor driver, and a power supply.

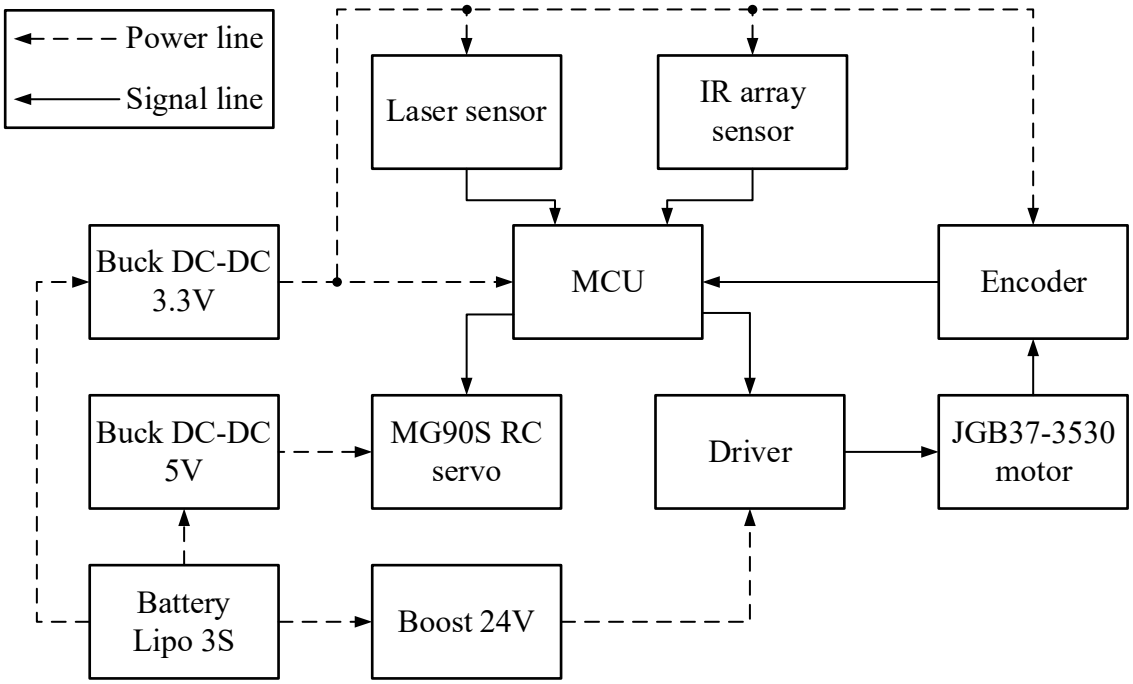


Figure 4.1. Overall electrical system diagram

The MCU receives signals from the sensors and encoder, executes the control algorithms, and generates commands for the steering actuator and rear-wheel motor.

The power system is divided into two branches. The control branch supplies the MCU and sensors, while the actuator branch supplies the steering actuator and motor driver. Both branches are powered by the same battery through their respective voltage regulators.

4.2. Line following sensor design

The TCRT5000 infrared reflectance sensor is selected for line detection. The sensor array is designed to estimate the position of the black line on the white track and provide feedback for the line-tracking controller.

The design includes resistor selection, mounting height determination, sensor arrangement, circuit design, calibration, and measurement error evaluation.

4.2.1. Sensor specification and resistor selection

The TCRT5000 consists of an infrared LED emitter and a phototransistor receiver. The emitter illuminates the track surface, while the phototransistor generates a collector current according to the intensity of reflected light.

White surfaces generally reflect more infrared light than black surfaces, producing different sensor outputs.

The main specifications of the TCRT5000 are summarized in the below table:

Table 4.1. TCRT5000 specifications

Parameter	Specification
Operating range	$0.2 \div 15\text{mm}$
Overall diminsions	$10.2 \times 5.8 \times 7\text{mm}$
Emitter half-angle, α	$16°$
Receiver half-angle, β	$30°$
Emission wavelength	950nm
Maximum current through phototransistor	100mA
Maximum current through LED	60mA
Output current under standard conditions	1mA

The detection principle is based on light reflection: white surfaces reflect more infrared light, while black surfaces absorb it. The fixed spacing between the emitter and receiver ensures stable detection and consistent sensor alignment.

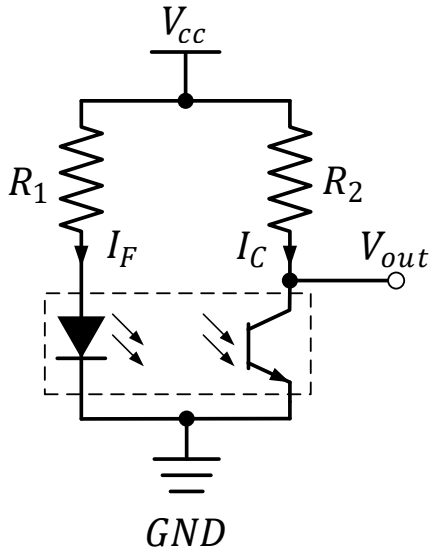


Figure 4.2. Operating principle of an infrared LED pair

With an operating voltage of $V_{CC} = 3.3\text{V}$, resistor $R_1$is connected in series with the infrared LED to limit its forward current. The current must be high enough to provide sufficient reflected light for detection while remaining within a safe operating range to extend the LED's lifespan.

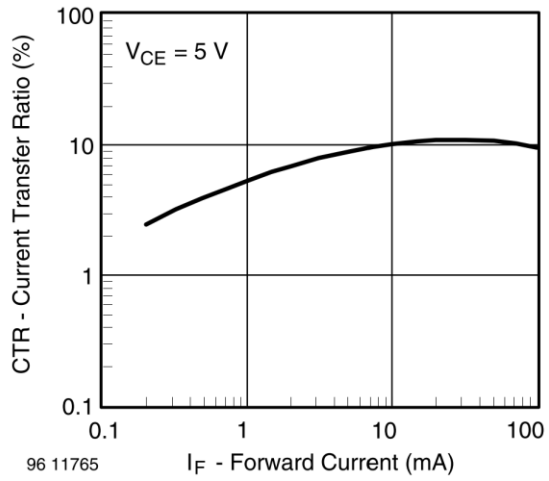


Figure 4.3. Relationship between current transfer ratio and forward current I_F[13]

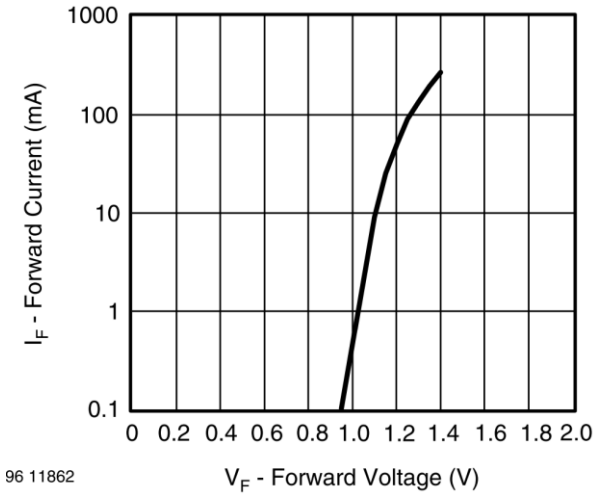


Figure 4.4. Forward current I_F versus forward voltage V_F[13]

According to Figure 4.3, a forward current (I_F) between 10 and 40 mA provides a current transfer ratio above 10%, which is the optimal current transfer ratio. To limit LED current and extend its operating life, $I_F = 10$mA is selected.

From the characteristic curve in Figure 4.4, the corresponding forward voltage is approximately $V_F = 1.1$V. Therefore, resistor $R_1$is calculated as follows:

$$R_1 = \frac{V_{cc} - V_F}{I_F} = \frac{3.3 - 1.1}{10 \times 10^{-3}} = 220\Omega \tag{4.1}$$

where, V_{cc}: is the supply voltage, $V_{cc} = 3.3$ (V).

V_F: is the LED forward voltage.

I_F: is the LED forward current.

Resistor R_2 converts the phototransistor collector current into an output voltage for the MCU.

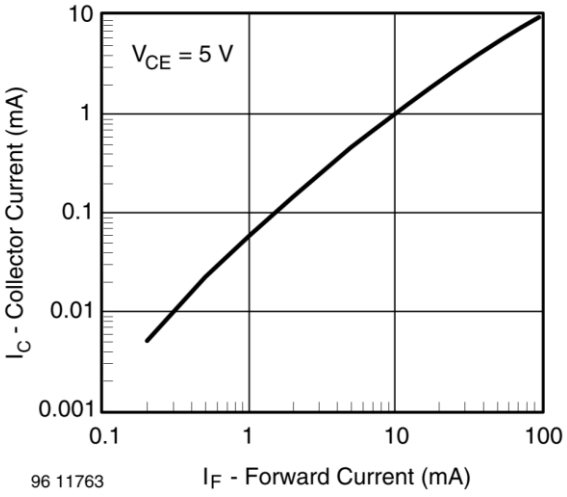


Figure 4.5. Relationship between current transfer ratio and forward current I_F[13]

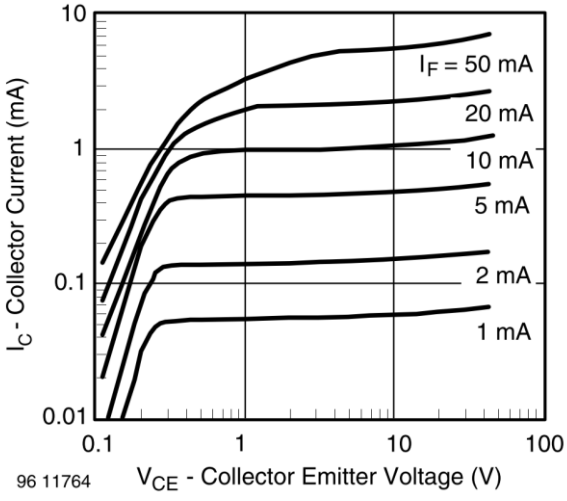


Figure 4.6. Collector emitter saturation voltage versus collector current[13]

Based on Figures 4.5 and 4.6, the collector current is approximately $I_C = 1\text{mA}$. Using the corresponding collector-emitter voltage of $V_{CE} = 1\text{V}$, the resistance is calculated as:

$$R_2 = \frac{V_{cc} - V_{CE}}{I_C} = \frac{3.3 - 1}{1 \times 10^{-3}} \approx 2.3\text{k}\Omega \tag{4.2}$$

The standard resistor value $R_2 = 2.4\text{k}\Omega$ is selected.

A capacitor of $C = 100\text{nF}$ is added to form a first-order RC low-pass filter with R_2, reducing high-frequency noise in the sensor output. The cutoff frequency is estimated by:

$$\begin{aligned} f_c &= \frac{1}{2\pi R_2 C} \\ &= \frac{1}{2\pi \times (2.4 \times 10^3) \times 100 \times 10^{-9}} \approx 663\ (\text{Hz}) \end{aligned} \tag{4.3}$$

The calculated cutoff frequency is approximately 660 Hz, allowing the filter to attenuate high-frequency noise from switching components and PWM signals while preserving the line sensor output. Therefore, a 100 nF capacitor is selected. Its effect on signal delay and distortion will be verified experimentally at the robot's operating speed.

4.2.1. Sensor orientation

According to the datasheet[14], the collector current I_C decreases from I_{C1} to I_{C2} as the sensor moves from a bright to a dark surface. The switching distance X_d is defined as the displacement over which the collector current changes I_C between 10% and 90% of its transition range. A smaller X_d indicates higher spatial resolution and better line-edge detection.

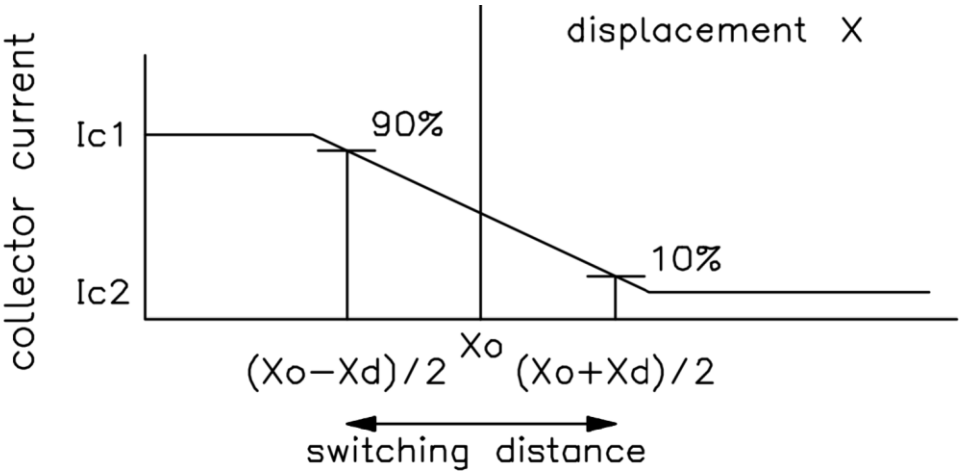


Figure 4.7. Abrupt reflection change with associated I_C curve[14]

Two sensor orientations are considered, as illustrated in below figure. In Position 1, the emitter–receiver axis is perpendicular to the black–white boundary, whereas in Position 2, it is parallel to the boundary.

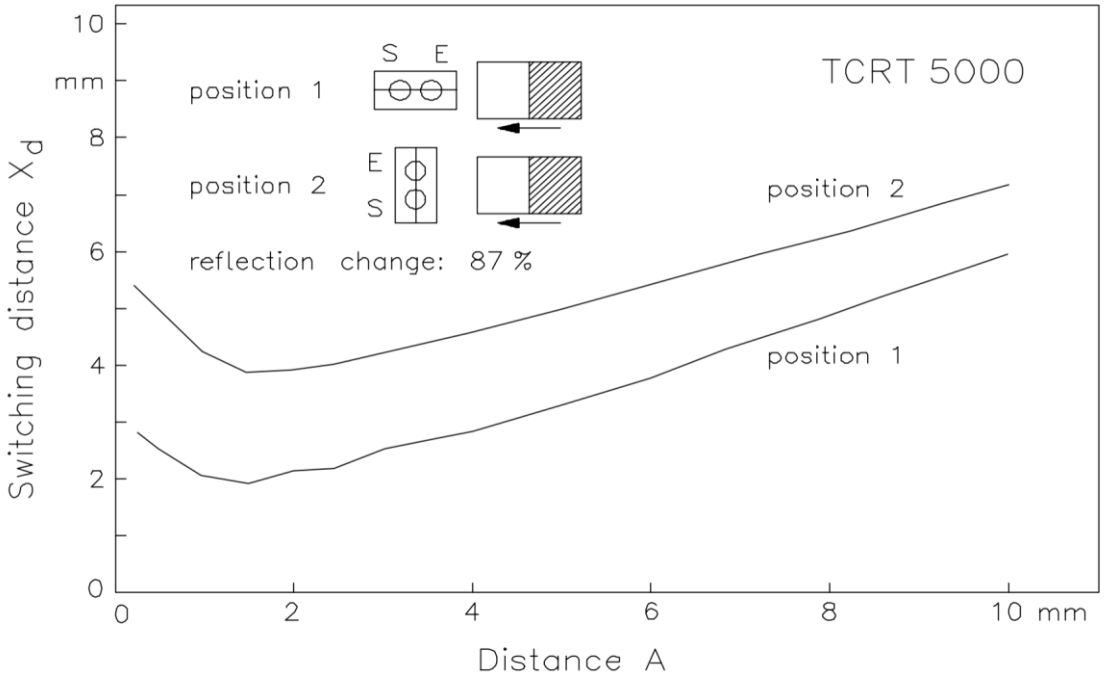


Figure 4.8. Comparison of two sensor orientations[14]

At the same mounting distance, Position 2 has a larger switching distance; therefore, its individual edge resolution is lower. However, the wider transition region provides more intermediate analog values, which is suitable for continuous line-position estimation. In addition, Position 2 places the 5.8 mm side of the TCRT5000 along the sensor-array direction, allowing a smaller sensor spacing.

4.2.2. Height sensor and calibration

For Position 2, the emitter–receiver axis is parallel to the black–white boundary. Therefore, the effective sensing region is evaluated by the overlap width y_d in the lateral direction.

A simplified geometric model is adopted in which the emission and reception regions are approximated as two circular footprints. The design criterion is chosen as $y_d > 3$mm, where 3mm is selected to provide an optical transition region comparable to the allowable line-following error.

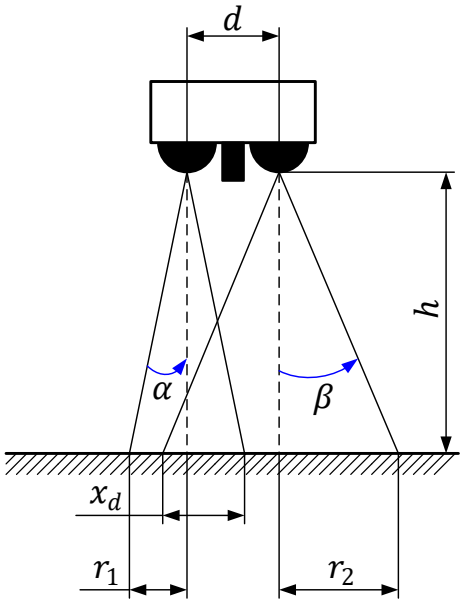


Figure 4.9. The emission and reception angles of the TCRT5000 sensor

The infrared LED and phototransistor each have fixed emission and reception angles, which determine the ground coverage at a mounting height h. The distance between emitter and collector is $d \approx 3.6\text{mm}$. With $\alpha = 16°$ and $\beta = 30°$, the lateral spreads are:

$$\begin{cases} r_1 = h \tan \alpha = h \tan 16° \\ r_2 = h \tan \beta = h \tan 30° \end{cases} \tag{4.4}$$

Case 1: No overlap

The two optical regions do not intersect when:

$$d \geq r_1 + r_2 \tag{4.5}$$

In this case, $y_d = 0\text{mm}$ so the design is not satisfied.

Case 2: Partial overlap

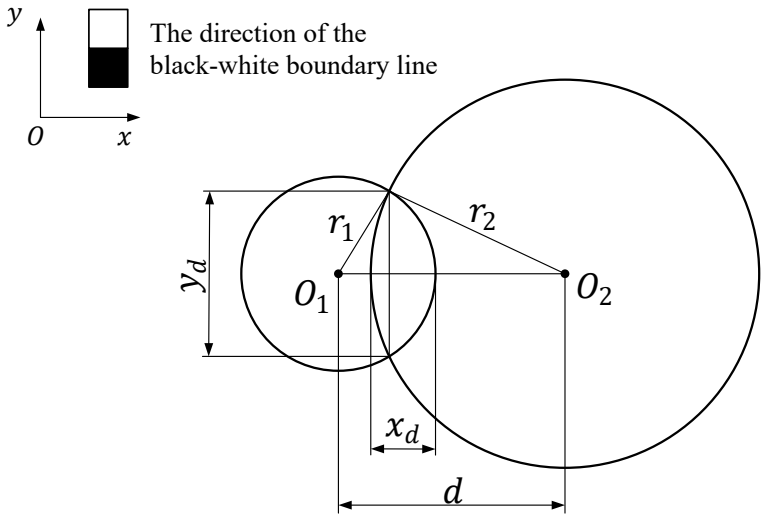


Figure 4.10. Partial overlap

When the two circles intersect and the common chord lies in front of O_1:

$$\begin{cases} r_1 + r_2 > d \\ r_2^2 < r_1^2 + d^2 \end{cases} \quad (4.6)$$

$$=> 4.17\ (\text{mm}) < h < 7.18\ (\text{mm})$$

The overlap width is:

$$y_d = \frac{1}{d}\sqrt{4d^2 r_1^2 - (d_2 + r_1^2 - r_2^2)^2} > 3 \quad (4.7)$$

$$=> 5.72\ (\text{mm}) < h < 11.75\ (\text{mm})$$

Therefore, the valid range:

$$5.72\ (\text{mm}) < h < 7.18\ (\text{mm}) \quad (4.8)$$

Case 3: Large overlap

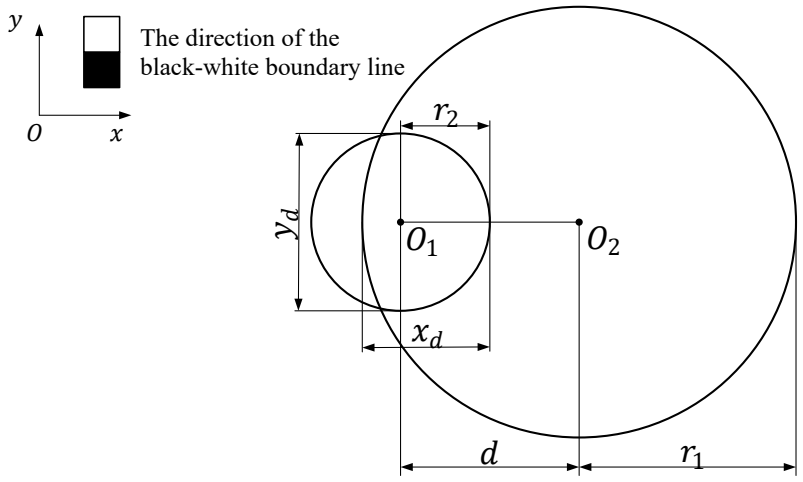


Figure 4.11. Large overlap

When the two circles intersect and the common chord lies in back of O_1:

$$r_2^2 \geq r_1^2 + d^2 \quad (4.9)$$

$$=> h \geq 7.18\ (\text{mm})$$

The overlap width is:

$$y_d = 2r_1 = 2h \tan 16° > 3 \quad (4.10)$$

$$=> h > 7.18\ (\text{mm})$$

According to the datasheet, the sensor operates most effectively within a distance range of $0.2 \div 15\text{mm}$. To identify the optimal mounting height, combined with case 2 and case 3, the ADC output was measured from 6 to 15mm above the surface. The results are shown in below table:

Table 4.2. ADC values from the line sensor at different distances

Height h (mm)	Balack value	White value	Difference
6	2696	336	2360
7	**2859**	**420**	**2439**
8	2987	621	2366
9	3105	954	2151
10	3215	1289	1926
11	3305	1556	1749
12	3390	1768	1622
13	3505	1968	1537
14	3574	2091	1483
15	3653	2269	1384

As the results from the above table, the height $h = 7\text{mm}$ is chosen due to the largest diffrence.

4.2.3. Distance between IR sensors

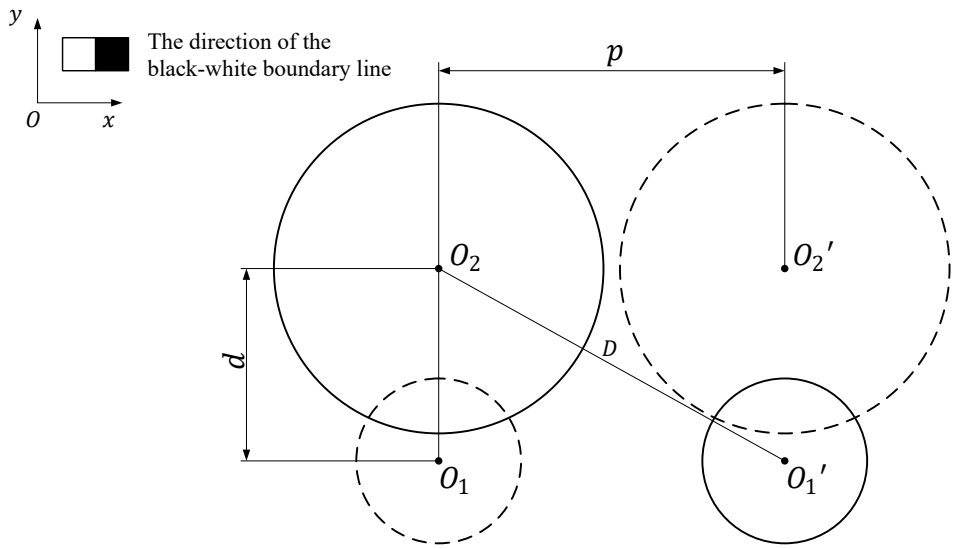


Figure 4.12. Space between two IR sensors

To avoid optical overlap between adjacent sensors, the spacing must be large enough to accommodate both spreads:

$$D^2 = d^2 + p^2 > (r_1 + r_2)^2 = h^2(\tan 16° + \tan 30°)^2 \quad (4.11)$$

$$=> p > 4.86 \text{ (mm)}$$

However, in Position 2, the 5.8 mm side of the TCRT5000 package is aligned with the sensor-array direction. Therefore, the sensor pitch must also satisfy the mechanical constraint $p \geq 5.8$ (mm).

The track width is $w = 26\text{mm}$. A pitch of

$$p = \frac{w}{2} = \frac{26}{2} = 13 \text{ (mm)} \tag{4.12}$$

is selected. When the robot is centered on the line, the two line boundaries are located at $x = 13\text{mm}$, directly below the sensors adjacent to the center sensor. This arrangement provides high sensitivity to small lateral deviations around the desired tracking position while providing sufficient spacing between adjacent sensors.

4.2.4. Quantiy of IR sensors in a row

With this spacing, the minimum number of sensors needed to cover the 26 mm line width:

$$n_{min} = \frac{26}{13} + 1 = 3 \tag{4.13}$$

An odd number provides a clear center reference for line tracking, so seven TCRT5000 sensors are selected to provide sufficient sensing width for line tracking and recovery.

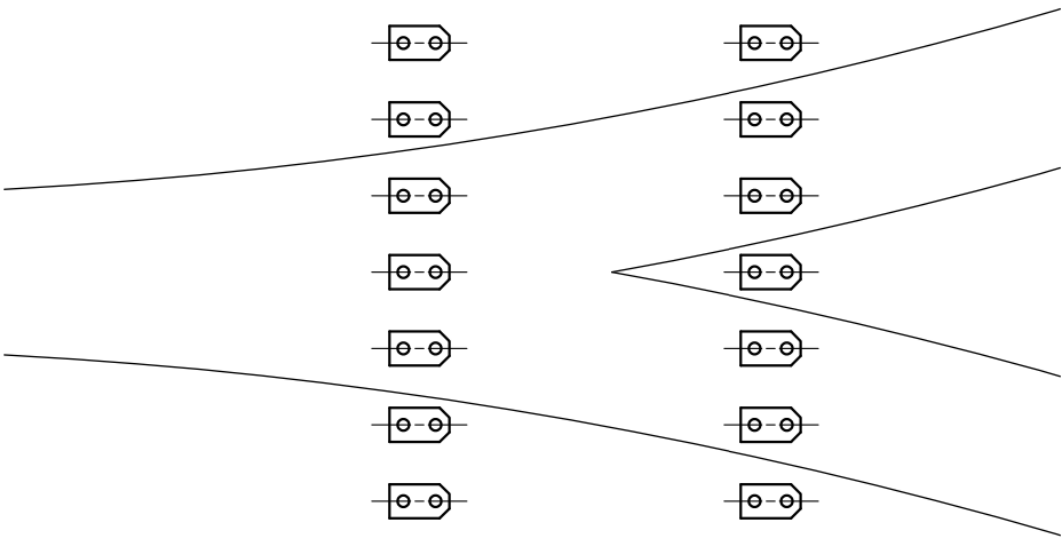


Figure 4.13. Position of sensors relative to the line

With a pitch of $p = 13\text{mm}$, the sensor positions are

$$x_i = [-39, -26, -13, 0, 13, 26, 39] \text{ (mm)} \tag{4.14}$$

The total sensor-array span is:

$$L = (7 - 1) \times 13 = 78 \text{ (mm)} \tag{4.15}$$

which provides a line-detection margin of

$$e_r = \frac{78 - 26}{2} = 26\ (\text{mm}) \tag{4.16}$$

Therefore, a seven-sensor array is selected.

4.2.5. Sensor circuit schematic design

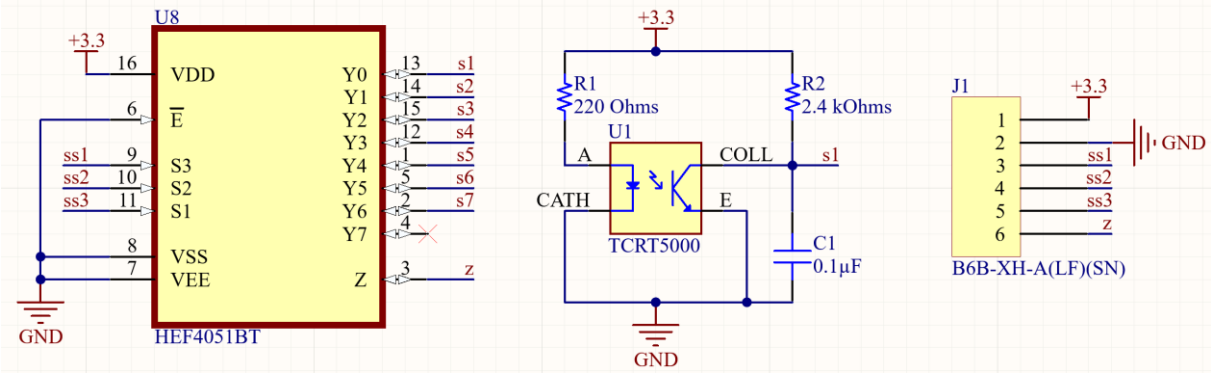


Figure 4.14. Schematic diagram of the line sensor circuit

Above figure shows the TCRT5000 sensor interface. The seven analog sensor outputs are connected to an HEF4051B multiplexer and sequentially selected by the ESP32 through three selection signals. The multiplexed output is then read by a single ADC channel.

4.2.6. Calibration and linearization of sensor ouputs

4.2.7. Calibration validation and error analysis

4.2.8. Error estimation using the weighted average method

4.3. Power supply design

The power supply system must provide the required voltage and current for all electrical components while maintaining stable operation of the control and actuator subsystems. Therefore, the power requirements of each subsystem are first determined before selecting the battery and voltage converters.

The electrical system can be divided into three main power branches:

Table 4.3. Three main power branches of electrical system

Subsystem	Required voltage	Main components
Driving system	24V	JGB37-3530 motor and driver
Steering system	5V	MG90S Servo
Control and sensing system	3.3V	MCU, TCRT5000, encoder, VL53L0X

4.3.1. Power requirement

4.3.1.1. Driving motor

The selected JGB37-3530-24100 DC motor operates at a rated voltage of 24 V and a rated current of 0.63 A. Therefore, its rated electrical input power is:

$$P_{motor,rated} = V_{motor} I_{rated} \tag{4.17}$$
$$= 24 \times 0.63 = 15.12 \text{ (W)}$$

where, V_{motor}: is the supply voltage of the motor, $V_{motor} = 24$ (V).

I_{rated}: is the rated current of the motor, $I_{rated} = 0.63$(A).

For power-supply sizing, the stall condition is also considered. With a stall current of approximately 3 A, the maximum electrical power demand is:

$$P_{motor,max} = 24 \times 3 = 72 \text{ (W)}$$

4.3.1.2. Line sensor array

Seven TCRT5000 sensors are used. In the previous sensor design, the IR LED current is selected as $I_F = 10$mA, with $R_1 = 220\Omega$, while the phototransistor pull-up resistor is $R_2 = 2.4k\Omega$.

The maximum current through the pull-up branch is approximately:

$$I_C \leq \frac{3.3}{2400} = 1.375 \text{ (mA)}$$

Hence, the maximum current of one sensor is approximately:

$$I_{sensor} \approx 10 + 1.375 = 11.375 \text{ (mA)}$$

For seven sensors:

$$I_{IR} = 7 \times 11.375 \approx 79.6 \text{ (mA)}$$

and:

$$P_{IR} = 3.3 \times 0.0796 \approx 0.263 \text{ (W)}$$

4.3.1.3. Control and sensing branch

The VL53L0X consumes approximately 19 mA during active ranging, with a peak current of up to approximately 40 mA according to the manufacturer datasheet[15]. The corresponding design power is:

$$P_{Laser} = 3.3 \times 0.04 = 0.132 \text{ (W)}$$

The ESP32-WROOM-32E operates from a 3.3 V supply. Espressif specifies an operating supply of 3.0–3.6 V and requires the external power source to be capable of supplying at least 0.5 A [16]. The corresponding design power is:

$$P_{ESP} = 3.3 \times 0.5 = 1.65 \text{ (W)}$$

4.3.1.4. Steering actuator

The MG90S servo is supplied from the 5 V rail. For power-supply design, a maximum current of approximately 2 A is assumed under stall or high-load conditions.

$$P_{servo,max} = 5 \times 2 = 10 \text{ (W)}$$

is selected to provide sufficient transient current margin.

4.3.1.5. Required power for each rail

3.3v rail

$$P_{3.3} = P_{ESP} + P_{Laser} + P_{IR}$$
$$= 1.65 + 0.132 + 0.263 = 2.045 \text{ (W)}$$

5v rail

$$P_5 = P_{servo} = 10(W)$$

24v rail

$$P_{24,max} = P_{motor,max} = 72 \text{ (W)}$$

4.3.2. Battery voltage selection

A 3S lipo battery is selected as the main power source. Its nominal voltage is:

$$V_{bat,nom} = 3 \times 3.7 = 11.1 \text{ (V)}$$

and the fully charged voltage is:

$$V_{bat,max} = 3 \times 4.2 = 12.6 \text{ (V)}$$

For power-converter sizing, the minimum design battery voltage is assumed to be approximately:

$$V_{bat,min} = 10 \text{ (V)}$$

Therefore, the power system is designed for an input-voltage range of approximately $10 - 12.6$V. The 3S battery voltage is directly stepped down to 5V and 3.3V, while the 24V motor rail is generated by a boost converter.

4.3.3. Battery capacity and discharge capability

For conservative sizing, the 3.3V rail is designed for 1A:

$$P_{3.3,design} = 3.3 \times 1 = 3.3 \text{ (W)}$$

Assuming a converter efficiency of 90%, the maximum battery current at $V_{bat,min} = 10$ (V) is

$$I_{bat,max} = \frac{72 + 10 + 3.3}{10 \times 0.9} \approx 9.48 \text{ (A)} \quad (4.18)$$

For a 2200mAh battery:

$$C_{required} = \frac{9.48}{2.2} \approx 4.31 \text{ (C)} \quad (4.19)$$

Therefore, a 3S 2200 mAh LiPo battery with a discharge rating of at least 5C is sufficient. A 35C battery is selected to provide additional current margin.

4.3.4. DC – DC converter selection

Two LM2596 fixed-output buck converters are used for 5V rail and 3.3V rail.

Based on the LM2596 datasheet[17] and the selected operating conditions, the same external component set is used for both converters:

$$L = 47\mu H, \quad D = SS54, \quad C_{in} = 470\mu F/50V, \quad C_{out} = 330\mu F/35V$$

The selected inductor current rating is at least 3 A, while the capacitors are low-ESR types suitable for switching-regulator operation. Using the same external components for both converters simplifies the PCB layout and component selection.

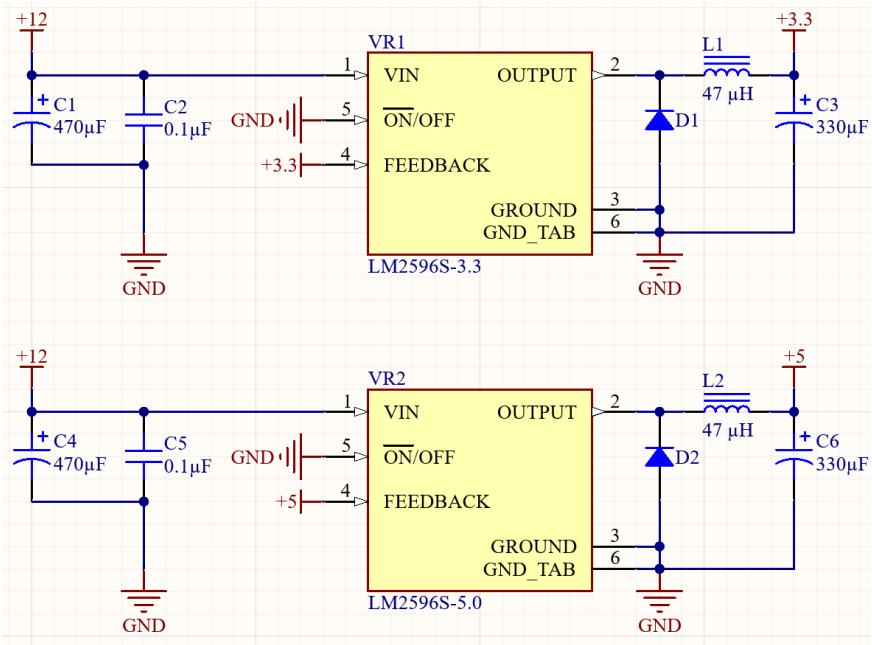


Figure 4.15. Schematic diagram of two DC – DC buck LM2596 in 3.3V (above) rail and 5V rail (below)

4.3.4.1. 24V boost converter

The motor branch requires a peak power of $P_{motor,max} = 72W$.

At $V_{bat,min} = 10V$ and $\eta = 90\%$:

$$I_{bat,max} = \frac{72}{10 \times 0.9} = 8\ (A) \tag{4.20}$$

Therefore, the selected boost converter must satisfy approximately:

$$V_{in} \leq 10V, \quad V_{out} = 24V, \quad P_{out} \geq 72W, \quad I_{in} \geq 8A$$

4.4. Complete system circuit design

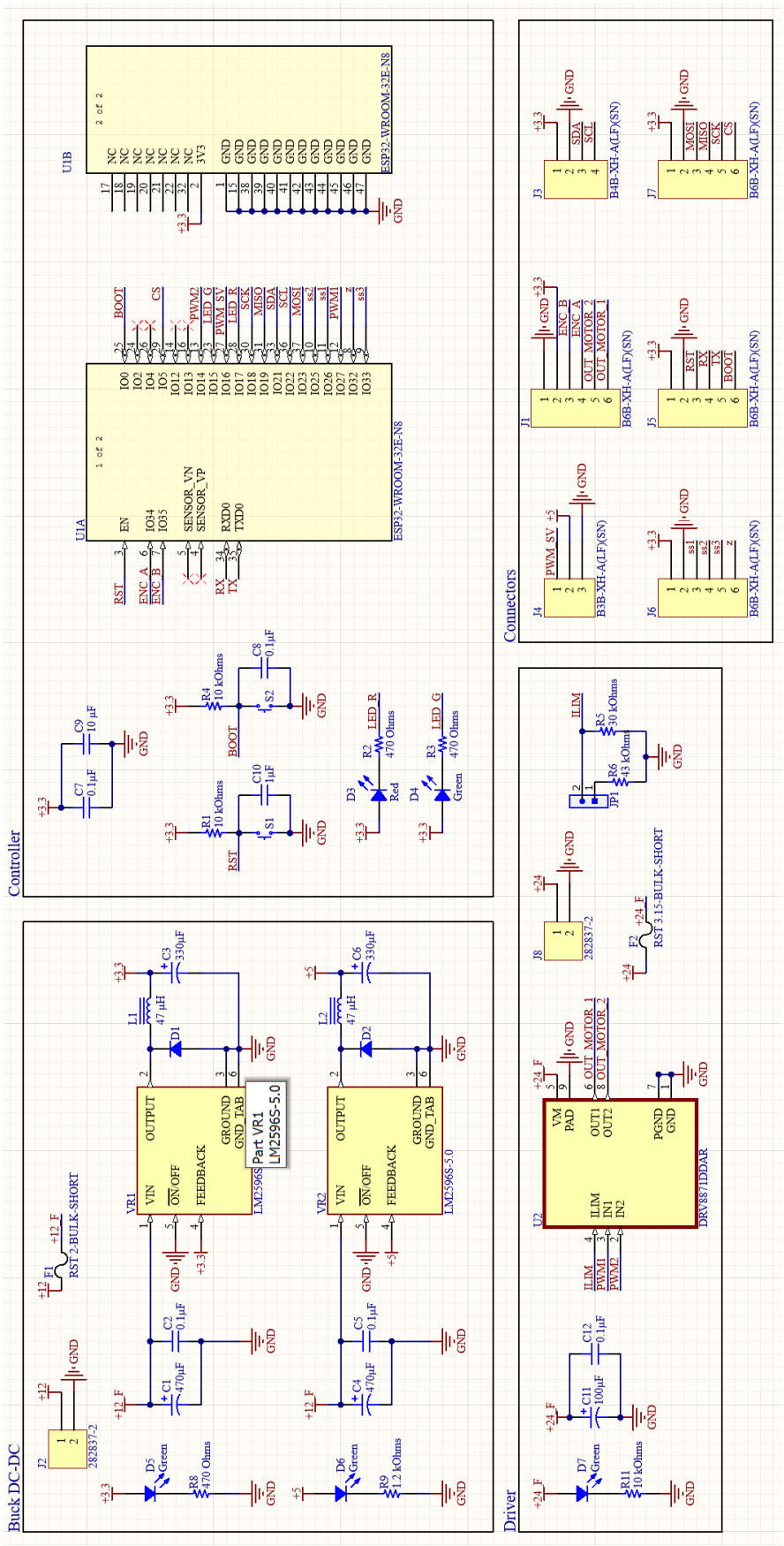


Figure 4.16. Schematic diagram of the controller circuit

Above figure shows the complete electrical schematic, integrating the ESP32-WROOM-32E, sensor interfaces, LM2596 power converters, DRV8871 motor driver, servo, encoder, and external connectors. The system is divided into 3.3 V, 5 V, and 24 V power branches for the control, steering, and driving subsystems, respectively.

Table 4.4. Summary of selected electrical components

Subsystem	Selected component	Supply
Main controller	ESP32 – WROOM – 32E	3.3V
Line sensor	7 × TCRT5000	3.3V
Analog multiplexer	HEF4051B	3.3V
Obstacle sensor	VL53L0X	3.3V
Steering actuator	MG90S	5V
Drive motor	JGB37 - 3530	24V
Motor driver	DRV8871	24V
Buck converter	LM2596S-3.3	3.3V
Buck converter	LM2596S-5.0	5V
Boost converter	400 W DC–DC boost module	24V output
Battery	3S LiPo, 2200 mAh, 35C	12V ouput

CHAPTER 5. CONTROLLER DESIGN AND MODELING

5.1. System modeling

The objective of this project is to develop a control system that enables the robot to follow a predefined line. Under ideal operating conditions, with the robot moving on a planar surface and carrying a constant load, complex dynamic effects can be neglected, allowing the analysis to focus primarily on the robot's kinematic behavior.

The robot is designed with a single front steering wheel and two rear driven wheels connected through a differential mechanism. Due to the mechanical characteristics of the differential, the rotational speeds of the two rear wheels are automatically distributed to accommodate differences in wheel travel during cornering. Therefore, the two rear wheels can be equivalently represented by a virtual wheel located at the center of the rear axle.

The main geometric parameters and reference points of the model are defined as follows:

Table 5.1. The main geometric parameters and reference points of the mode

Symbol	Parameters	Symbol	Parameters
A	Center of rear wheel axis	l_2	Length to sensor location
B	Center of steering shaft	ω	Angular velocity of robot
C	Center of sensor	D	Radius of wheel
l_1	Length to steering shaft	v	Linear velocity at point A
R	Reference point	e_1	Vertical error
e_2	Horizontal error	e_3	Angular error of robot heading to line direction
v_d	Desired velocity	–	–

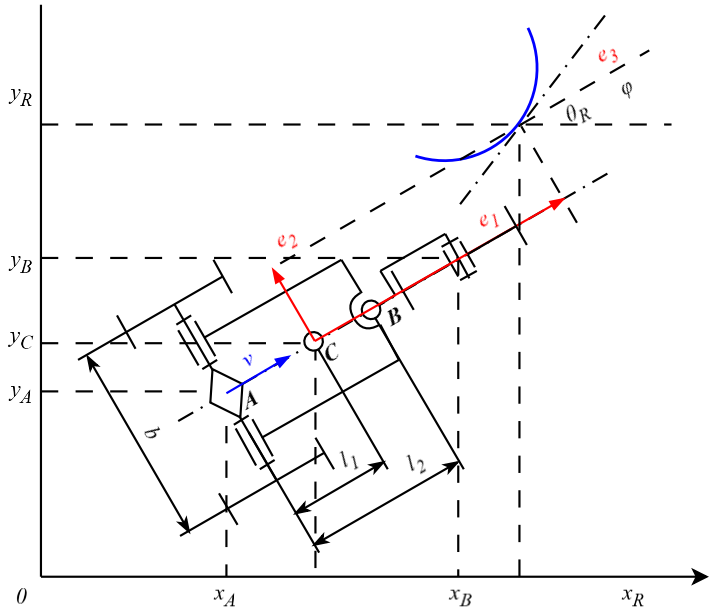


Figure 5.1: Kinematic model coordinate system

5.1.1. Kinematic equations

Consider the global coordinate system Oxy. The kinematic equations describing the velocity of the robot at point A are given by:

$$\begin{cases} \dot{x}_A = v\cos\theta \\ \dot{y}_A = v\sin\theta \end{cases} \quad (5.1)$$

The position of point C, where the line-tracking sensors are installed, in the global coordinate system is determined from the coordinates of point A and the distance l_2:

$$\begin{cases} x_C = x_A + l_2\cos\theta \\ y_C = y_A + l_2\sin\theta \end{cases} \quad (5.2)$$

Taking the time derivative of both sides yields the velocity components at point C:

$$\begin{cases} \dot{x}_C = \dot{x}_A - l_2\dot{\theta}\sin\theta \\ \dot{y}_C = \dot{y}_A + l_2\dot{\theta}\cos\theta \end{cases} \quad (5.3)$$

Substituting the velocity equations at point A and noting that $\dot{\theta} = \omega$, the kinematic equations at point C become:

$$\begin{cases} \dot{x}_C = v\cos\theta - l_2\dot{\theta}\sin\theta \\ \dot{y}_C = v\sin\theta + l_2\dot{\theta}\cos\theta \\ \dot{\theta}_C = \dot{\theta} = \omega \end{cases} \quad (5.4)$$

5.1.2. Error equations

Let R denote the desired reference point located on the line to be tracked. The desired linear velocity and angular velocity at this reference point are denoted by v_d and ω_d, respectively. The reference-point kinematics are expressed as:

$$\begin{cases} \dot{x}_R = v_d \cos\theta_R \\ \dot{y}_R = v_d \sin\theta_R \\ \dot{\theta}_R = \omega_d \end{cases} \quad (5.5)$$

The positional and heading errors between the reference point R and the actual sensor point C are defined in the global coordinate system as:

$$\begin{cases} e_x = x_R - x_C \\ e_y = y_R - y_C \\ e_\theta = \theta_R - \theta \end{cases} \quad (5.6)$$

Taking the derivative of (5.6):

$$\begin{cases} \dot{e}_x = \dot{x}_R - \dot{x}_C \\ \dot{e}_y = \dot{y}_R - \dot{y}_C \\ \dot{e}_\theta = \dot{\theta_R} - \dot{\theta} \end{cases} \quad (5.7)$$

Substituting (5.4) into (5.7):

$$\begin{cases} \dot{e}_x = \dot{x}_R - v\cos\theta + l_2\dot{\theta}\sin\theta \\ \dot{e}_y = \dot{y}_R - v\sin\theta - l_2\dot{\theta}\cos\theta \\ \dot{e}_\theta = \dot{\theta_R} - \dot{\theta} \end{cases} \quad (5.8)$$

5.1.3. Transformation to body frame

The position and orientation errors (e_x, e_y, e_θ)defined in Section 5.1.2 are expressed in the global coordinate frame Oxy. However, to allow the robot to adjust its steering angle and velocity based on the sensor measurements, these errors must be expressed relative to the robot's current direction of motion, namely, in the robot body frame.

To transform the position error from the global frame Oxy to the robot body frame, the error vector $\overrightarrow{CR} = [x_R - x_C \quad y_R - y_C]^T$ is projected onto the local coordinate axes of the robot. The robot orientation with respect to the global Ox-axis is denoted by θ. Therefore, the error vector is multiplied by the rotation matrix $R(-\theta)$.

The transformed error variables are given by:

$$\begin{cases} e_1 = (x_R - x_C)\cos\theta + (y_R - y_C)\sin\theta \\ e_2 = -(x_R - x_C)\sin\theta + (y_R - y_C)\cos\theta \\ e_3 = \theta_R - \theta \end{cases} \quad (5.9)$$

In matrix form, the error transformation is expressed as:

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_R - x_C \\ y_R - y_C \\ \theta_R - \theta \end{bmatrix} \quad (5.10)$$

This transformation expresses the tracking errors relative to the robot's own orientation, which is more suitable for designing the trajectory-tracking controller.

5.1.4. Final error dynamics

To design the control law, the relationship between the time derivatives of the body-frame errors $(\dot{e}_1, \dot{e}_2, \dot{e}_3)$ and the velocity inputs (v, ω) must be established. This is achieved by differentiating the error equations defined in Section 5.1.3 with respect to time.

$$\begin{cases} \dot{e}_1 = \dot{e}_x \cos\theta - e_x\dot{\theta}\sin\theta + \dot{e}_y \sin\theta + e_y\dot{\theta}\cos\theta \\ \dot{e}_2 = -\dot{e}_x \sin\theta - e_x\dot{\theta}\cos\theta + \dot{e}_y \cos\theta - e_y\dot{\theta}\sin\theta \\ \dot{e}_3 = \dot{\theta}_R - \dot{\theta} \end{cases} \tag{5.11}$$

Grouping the terms containing $\dot{\theta}$, where $\dot{\theta} = \omega$, and substituting the definitions of e_1 and e_2, the above equations become:

$$\begin{cases} \dot{e}_1 = \left(\dot{e}_x \cos\theta + \dot{e}_y \sin\theta\right) + \omega e_2 \\ \dot{e}_2 = \left(-\dot{e}_x \sin\theta + \dot{e}_y \cos\theta\right) - \omega e_1 \\ \dot{e}_3 = \omega_d - \omega \end{cases} \tag{5.12}$$

Next, the error dynamics obtained in Section 5.1.2 are substituted into the two grouped terms.

Derivation of $\dot{e}_1$

$$\dot{e}_x \cos\theta + \dot{e}_y \sin\theta = \begin{matrix} [v_d \cos\theta_R - (v\cos\theta - l_2\omega\sin\theta)]\cos\theta \\ +[v_d \sin\theta_R - (v\sin\theta + l_2\omega\cos\theta)]\sin\theta \end{matrix} \tag{5.13}$$

$$\Longrightarrow \dot{e}_x \cos\theta + \dot{e}_y \sin\theta = v_d \cos(\theta_R - \theta) - v \tag{5.14}$$

Derivation of $\dot{e}_2$

$$-\dot{e}_x \sin\theta + \dot{e}_y \cos\theta = \begin{matrix} -[v_d \cos\theta_R - v\cos\theta + l_2\omega\sin\theta]\sin\theta \\ +[v_d \sin\theta_R - v\sin\theta - l_2\omega\cos\theta]\cos\theta \end{matrix} \tag{5.15}$$

$$\Longrightarrow -\dot{e}_x \sin\theta + \dot{e}_y \cos\theta = v_d \sin(\theta_R - \theta) - l_2\omega \tag{5.16}$$

Substituting the simplified expressions into the body-frame error equations yields the final error dynamics:

$$\begin{cases} \dot{e}_1 = v_d \cos e_3 - v + \omega e_2 \\ \dot{e}_2 = v_d \sin e_3 - \omega(l_2 + e_2) \\ \dot{e}_3 = \omega_d - \omega \end{cases} \tag{5.17}$$

5.1.5. Summary of the system model

In summary, the kinematic model used for the line-following robot controller design is characterized by two fundamental mathematical representations.

Error Transformation Matrix:

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_R - x_C \\ y_R - y_C \\ \theta_R - \theta \end{bmatrix} \tag{5.18}$$

Error Dynamics Matrix:

$$\begin{bmatrix}\dot{e}_1\\ \dot{e}_2\\ \dot{e}_3\end{bmatrix} = \begin{bmatrix}v_d\, cos\, e_3\\ v_d\, sin\, e_3\\ \omega_d\end{bmatrix} + \begin{bmatrix}-1 & e_2\\ 0 & -(l_2+e_2)\\ 0 & -1\end{bmatrix}\begin{bmatrix}v\\ \omega\end{bmatrix} \quad (5.19)$$

The resulting error-dynamics model serves as the mathematical foundation for applying the direct Lyapunov method to synthesize the control law (v, ω). The objective is to regulate the robot's motion such that the tracking errors e_1, e_2, and e_3 converge toward zero, thereby enabling accurate trajectory tracking.

5.2. Controller design

5.2.1. Controller formulation

The kinematic feedback controller is developed based on the final error dynamics equations derived in Section 5.1:

$$\begin{cases}\dot{e}_1 = v_d\, cos\, e_3 - v + \omega e_2\\ \dot{e}_2 = v_d\, sin\, e_3 - \omega(l_2+e_2)\\ \dot{e}_3 = \omega_d - \omega\end{cases} \quad (5.20)$$

The control strategy relies on Lyapunov's direct method to find the appropriate control inputs $u = [v \quad \omega]^T$ that stabilize the system. We propose the following non-holonomic kinematic control law:

$$\begin{cases}v = v_d\, cos\, e_3 + k_1 e_1\\ \omega = \omega_d + k_2 v_d e_2 + k_3\, sin\, e_3\end{cases} \quad (5.21)$$

where $k_1, k_2, k_3 > 0$ are the positive control gains to be tuned.

5.2.2. Stability analysis

To mathematically prove the stability of the proposed controller, we select the following Lyapunov function candidate:

$$V = \frac{1}{2}(e_1^2 + e_2^2) + \frac{1 - cos\, e_3}{k_2} \quad (5.22)$$

This Lyapunov function is positive definite, satisfying $V \geq 0$ for all error states, and $V = 0$ if and only if $e_1 = e_2 = e_3 = 0$.

Taking the time derivative of V yields:

$$\dot{V} = e_1\dot{e}_1 + e_2\dot{e}_2 + \frac{sin\, e_3}{k_2}\dot{e}_3 \quad (5.23)$$

Substituting the error dynamics from the system model into the derivative:

$$\dot{V} = e_1(v_d\, cos\, e_3 - v + \omega e_2) + e_2\big(v_d\, sin\, e_3 - \omega(e_1 + l_2)\big) + \frac{sin\, e_3}{k_2}(\omega_d - \omega) \quad (5.24)$$

Expanding and regrouping the terms:

$$\dot{V} = e_1(v_d \cos e_3 - v) + e_2(v_d \sin e_3 - \omega l_2) + e_1\omega e_2 - e_2\omega e_1 + \frac{\sin e_3}{k_2}(\omega_d - \omega) \quad (5.25)$$

Notice that the cross-coupling terms $e_1\omega e_2$ and $-e_2\omega e_1$ perfectly cancel each other out, simplifying the equation to:

$$\dot{V} = e_1(v_d \cos e_3 - v) + e_2 v_d \sin e_3 - e_2\omega l_2 + \frac{\sin e_3}{k_2}(\omega_d - \omega) \quad (5.26)$$

Now, substituting the proposed control law $v = v_d \cos e_3 + k_1 e_1$ into the first term:

$$e_1\big(v_d \cos e_3 - (v_d \cos e_3 + k_1 e_1)\big) = -k_1 e_1^2 \quad (5.27)$$

Substituting the angular velocity control law $\omega = \omega_d + k_2 v_d e_2 + k_3 \sin e_3$ into the equation:

$$\dot{V} = -k_1 e_1^2 + e_2 v_d \sin e_3 - e_2 l_2 \omega + \frac{\sin e_3}{k_2}\big(\omega_d - (\omega_d + k_2 v_d e_2 + k_3 \sin e_3)\big) \quad (5.28)$$

$$\Longrightarrow \dot{V} = -k_1 e_1^2 + e_2 v_d \sin e_3 - e_2 l_2 \omega - e_2 v_d \sin e_3 - \frac{k_3}{k_2}\sin^2 e_3 \quad (5.29)$$

The terms $e_2 v_d \sin e_3$ and $-e_2 v_d \sin e_3$ gracefully cancel out, leaving:

$$\dot{V} = -k_1 e_1^2 - \frac{k_3}{k_2}\sin^2 e_3 - e_2 l_2 \omega \quad (5.30)$$

Finally, expanding the $-e_2 l_2 \omega$ term with the control law:

$$\dot{V} = -k_1 e_1^2 - \frac{k_3}{k_2}\sin^2 e_3 - e_2 l_2(\omega_d + k_2 v_d e_2 + k_3 \sin e_3) \quad (5.31)$$

$$\Longrightarrow \dot{V} = -k_1 e_1^2 - k_2 l_2 v_d e_2^2 - \frac{k_3}{k_2}\sin^2 e_3 - e_2 l_2(\omega_d + k_3 \sin e_3) \quad (5.32)$$

5.2.3. Stability conclusion

From the final expression of $\dot{V}$, we can observe that:

1. The terms $-k_1 e_1^2$, $-k_2 l_2 v_d e_2^2$, and $-\frac{k_3}{k_2}\sin^2 e_3$ are strictly negative, providing strong natural convergence for the tracking errors.
2. The presence of the sensor look-ahead distance l_2 introduces a damping effect ($-k_2 l_2 v_d e_2^2$), which actively improves the lateral stability of the robot.
3. The remaining term $-e_2 l_2(\omega_d + k_3 \sin e_3)$ represents a bounded disturbance. By selecting sufficiently large control gains for k_2 and k_3, the negative definite terms will heavily dominate.

Therefore, the proposed controller ensures that the system is stable in the sense of Lyapunov, and all tracking errors will converge to an arbitrarily small neighborhood around zero.

To prepare for the simulation and practical implementation, the preliminary controller gains are systematically selected. The units of the control parameters are scaled to accommodate millimeters and seconds, which are standard for the physical dimensions of the robot. Setting $k_1 = 1.5$ provides active compensation for the longitudinal tracking error, ensuring the robot constantly catches up to the virtual reference point. The steering response gains k_2 and k_3 are calibrated to stabilize the rapid lateral dynamics at high velocities. The preliminary gains are chosen as:

$$k_1 = 1.5, \quad k_2 = 0.00015, \quad k_3 = 10.0$$

5.2.4. Relationship between control inputs and actuator commands

To implement the computed control variables (v, ω) on the physical hardware, they must be converted into the steering angle (γ) and the driving motor speed (ω_m).

Steering angle calculation:

Let γ denote the steering angle of the front wheel, and v_B the linear velocity at the steering wheel center B. From the tricycle kinematic constraints:

$$v = v_B \cos\gamma$$

$$\omega = \frac{v_B \sin\gamma}{l_1}$$

Therefore, the required steering angle for the servo motor is:

$$\gamma = \arctan\left(\frac{l_1 \omega}{v}\right)$$

Motor speed command:

The required angular velocity of the rear drive motor (before the differential gearbox) is calculated as:

$$\omega_m = \frac{2v}{D} \cdot \frac{60}{2\pi} \cdot \eta$$

where: ω_m: motor speed in rpm

D: wheel diameter (m)

η: differential gear ratio

v: controlled linear velocity (m/s)

Final control relationships:

The following equations map the designed control inputs (v, ω) directly to the actual actuator commands (motor speed ω_m and steering angle γ) for hardware deployment:

$$\begin{cases} \omega_m = \dfrac{2v}{D} \cdot \dfrac{60}{2\pi} \cdot \eta \\ \gamma = \arctan\left(\dfrac{l_1 \omega}{v}\right) \end{cases}$$

5.3. Controller simulation

To evaluate the performance of the proposed controller and route-selection strategy, numerical simulations were performed using the previously established kinematic model. The track was divided into two scenarios according to the selected route, with the robot terminating on either the upper or lower branch.

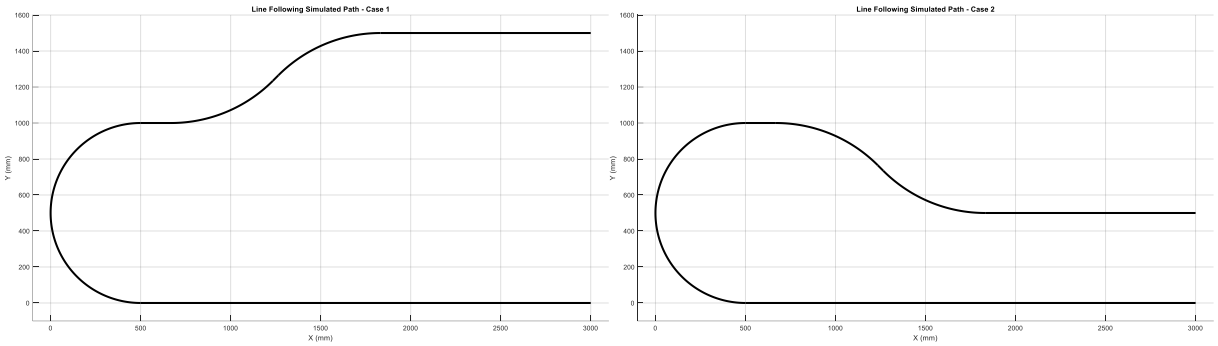


Figure 5.2. Simulated line-following path for case 1 and 2

For the trajectory-tracking evaluation, the reference point used for line following is the center of the line sensor array. Considering the 26 mm line width and the specified sensor positioning tolerance of ± 3 mm relative to both sides of the line, the allowable lateral deviation in the simulation is defined as ± 16 mm relative to the reference point on the robot.

An initial position deviation was introduced to evaluate the sensitivity of the controller. The proposed controller was then applied to both route-selection cases. The resulting reference trajectory and robot trajectory are presented in figure 5.4.

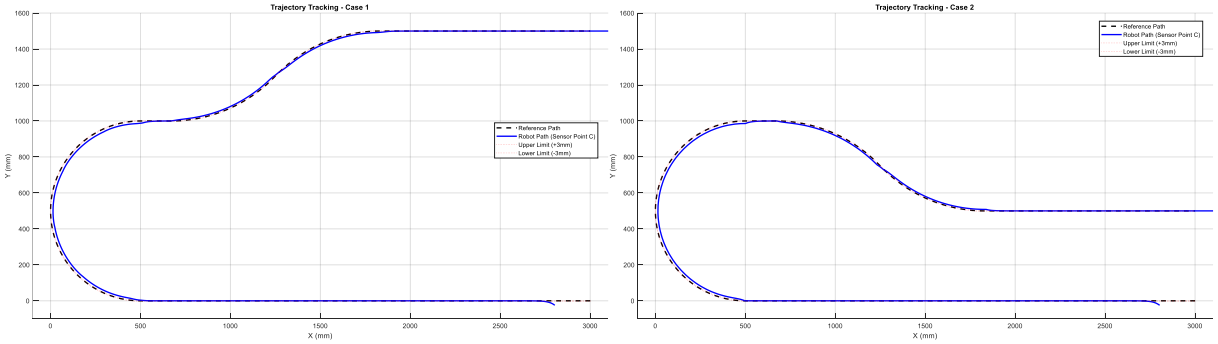
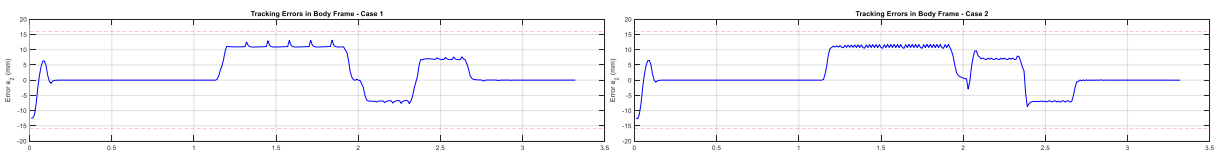


Figure 5.3. Trajectory tracking performance for case 1 and 2



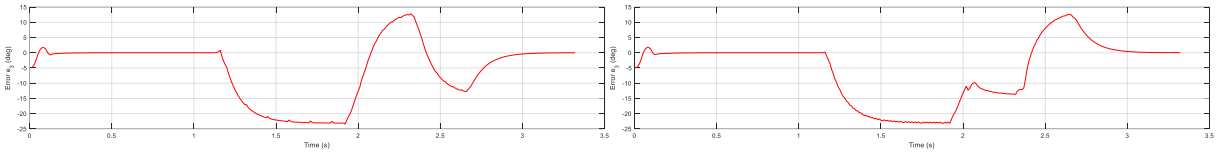


Figure 5.4. Lateral tracking error (e_2) and heading error (e_3) for case 1 and case 2

The simulation results show that the lateral and heading errors temporarily increase when the robot passes through sections with higher curvature. However, both errors subsequently decrease as the robot continues along the reference trajectory. This indicates that the controller can compensate for geometric variations in the path while maintaining stable motion.

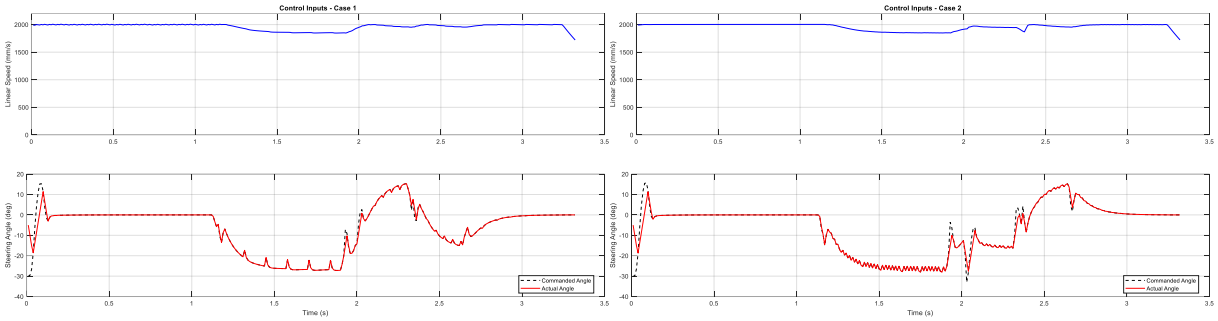


Figure 5.5. Linear velocity control input and steering angle control input for case 1 (left) and case 2 (right)

The control inputs are also evaluated for both cases. The linear velocity remains nearly constant, consistent with the constant-speed assumption used in the controller design. Meanwhile, the steering angle varies smoothly according to the trajectory curvature while remaining within the specified saturation limits.

Overall, the simulation results demonstrate that the proposed controller provides accurate and stable trajectory tracking for the considered vehicle model under different route geometries. Combined with the proactive obstacle-detection strategy, the simulation confirms the feasibility of implementing the proposed route-selection and trajectory-tracking method on the physical robot. Further controller tuning can be performed to improve tracking performance under practical operating conditions.

5.4. Proactive obstacle detection and route selection

5.4.1. VL53L0X sensor evaluation and calibration

The target obstacle was a white 3D-printed cylindrical object with a diameter of 50 mm and a height of 200 mm. According to the track geometry, the distance from the robot's centerline to the obstacle center is approximately 500 mm for the near branch and 1500 mm for the far branch. Considering the obstacle radius, the theoretical distance from the sensor to the reflecting surface of the cylinder for the near branch is approximately 475 mm.

Experimental results showed that the VL53L0X could reliably detect the narrow cylindrical obstacle at a frontal incidence angle only when the sensing distance was below approximately

700 mm. At a distance of 1500 mm, the small effective reflective cross-section combined with optical signal attenuation prevented reliable detection of the obstacle.

Therefore, the route-selection algorithm was simplified to a binary decision based only on the obstacle condition at the near branch, corresponding to a calibrated distance of approximately 475 mm.

Two route-selection cases are considered:

- Case 1 – Obstacle detected: If the measured distance indicates the presence of an obstacle within the 475 mm detection region, the near branch is considered blocked. The controller therefore sets the corresponding flag to select the upper route.
- Case 2 – No obstacle detected: If no obstacle is detected, with the measured value being outside the valid detection range or greater than approximately 700 mm, the obstacle is assumed to be located on the far branch. The controller therefore sets the corresponding flag to select the lower route.

5.4.2. Static data collection and noise analysis

To determine an appropriate detection threshold for the near branch, static measurements were performed with the center of the cylindrical obstacle positioned 500 mm from the sensor. The sensor data were collected at the specified sampling interval over a total of 600 samples.

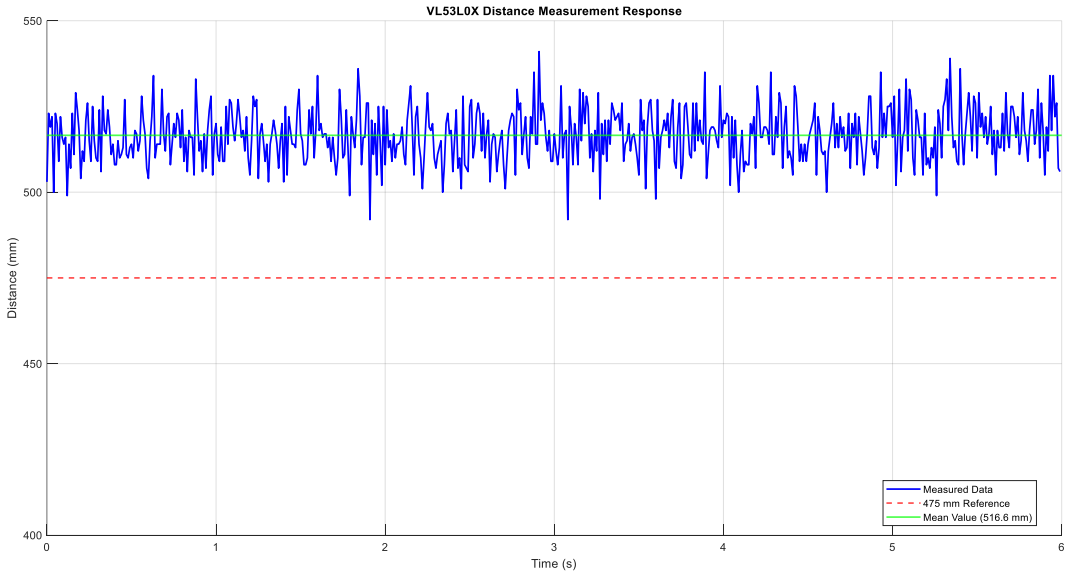


Figure 5.6. Raw VL53L0X distance measurement at 475 mm

Figure 5.1 shows the raw distance measurements obtained from the VL53L0X over time. The measured signal exhibits continuous fluctuations caused by random measurement noise. The mean measured distance is 516.6 mm, which differs from the theoretical reflecting-surface distance of 475 mm. This difference indicates the presence of a static measurement offset in the sensor output.

5.4.3. Sensor calibration and detection threshold

To compensate for the systematic measurement offset, a calibration offset was introduced into the signal-processing stage of the microcontroller.

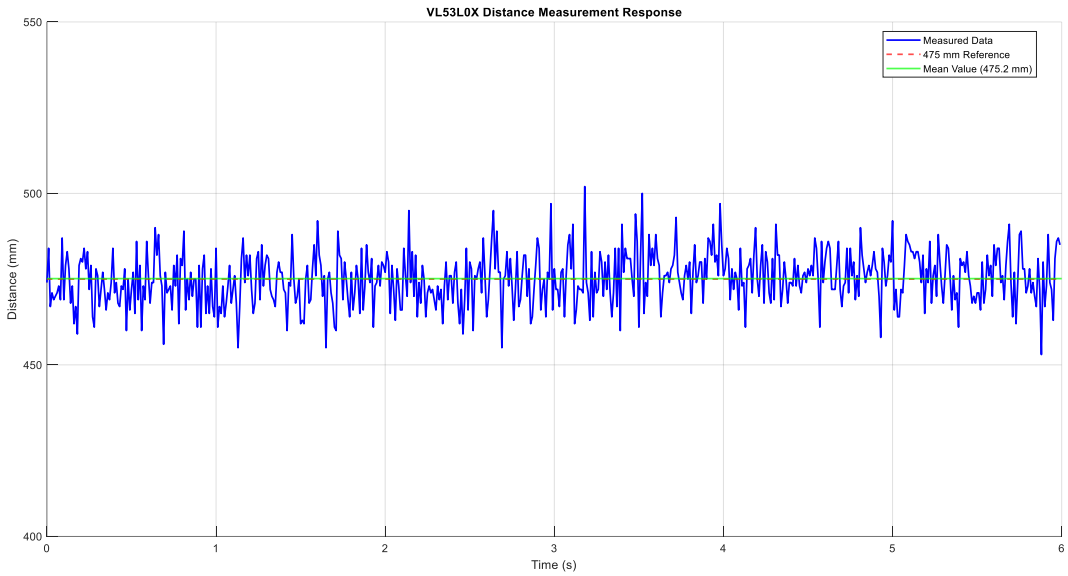


Figure 5.7. Calibrated VL53L0X distance measurement at 475 mm

After calibration, the mean measured distance decreased to 475.2 mm, which closely corresponds to the theoretical reference distance of 475 mm. As shown in Figure 5.2, the measurement signal still contains random fluctuations around the mean value; however, the center of the measured distribution is now aligned with the physical reflecting surface of the cylindrical obstacle.

5.5. Motor controller

Accurate motor modeling is essential for building a reliable speed controller. Instead of using nominal values from the datasheet, the motor's dynamic characteristics are determined experimentally with the motor mounted on the differential drive system.

5.5.1. Determine the transfer function of the motor system

A step input signal is applied to the motor via a PWM control signal, and the motor's speed response is measured by an encoder. This experiment characterizes the dynamic relationship between the input control signal and the motor's output speed, with the PWM signal incremented by 5 units every 1s, and a sampling time of 10ms.

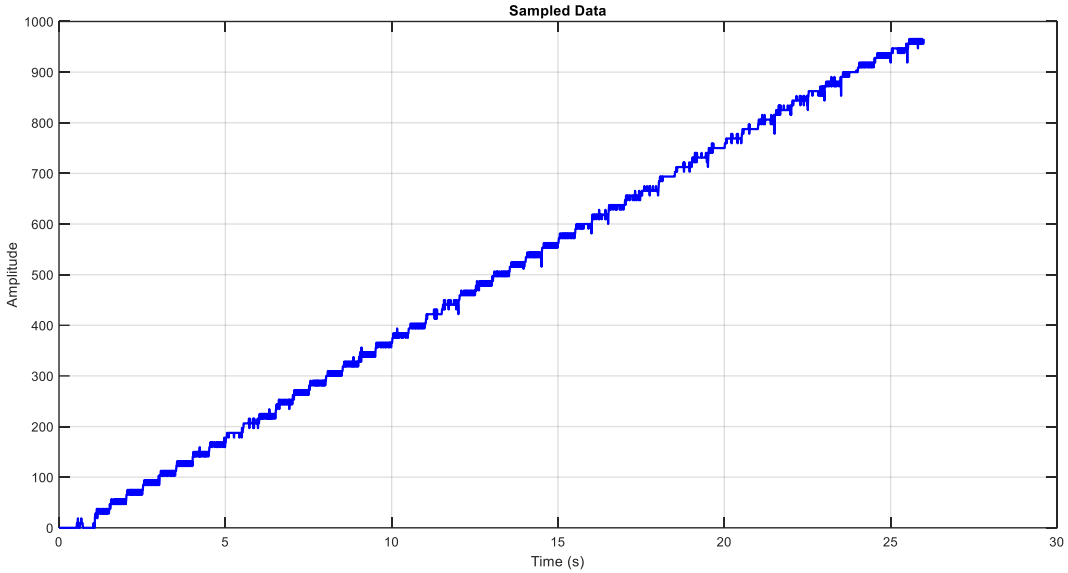


Figure 5.8. Experimental Response of JGB37-3530 speed under Incremental PWM Control

From the graph above, the deadband is determined to be in the PWM range from 0 to 15, in which no significant movement occurs despite the increase in the PWM input signal. Beyond this range, the motor speed increases almost linearly with the input PWM signal.

To determine the transfer function of the motor system, the input signal is applied as a step and maintained at a stable input level while measuring the motor speed. The experiment is conducted with the following main parameters:

- Sampling period of 10ms. The desired sampling period when the vehicle is operating.
- Fixed PWM selected is 190. According to the survey above, it is in the range of 700-750 RPM, the desired operating range of the robot.

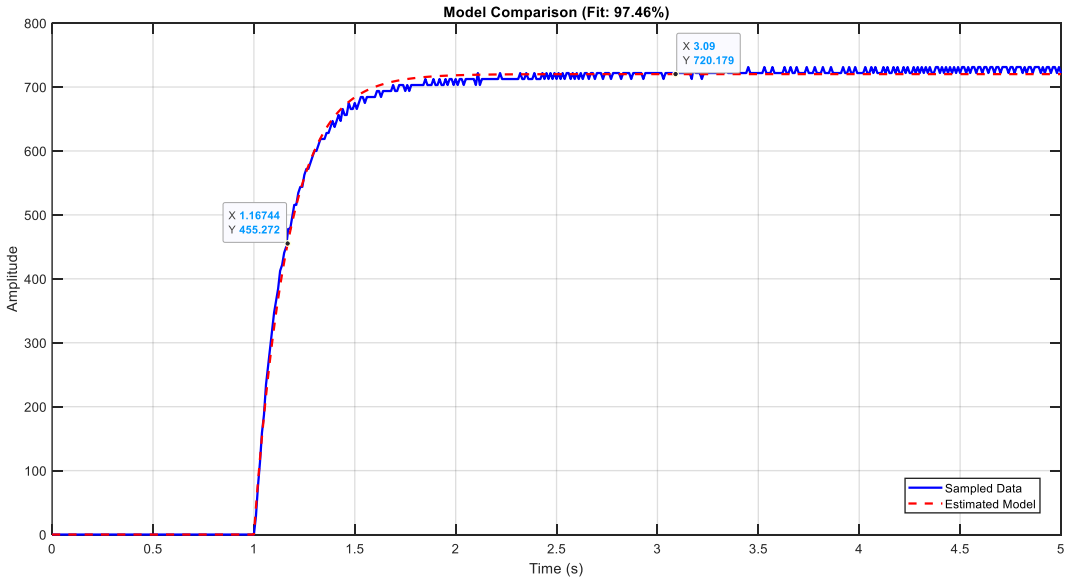


Figure 5.9. Measured motor speed response to a step input

From the measured response, the steady-state speed is reached at approximately $0.5s$. The motor speed stabilizes when the response is around $720RPM$; therefore, 63.2% of the steady-state speed is $455RPM$. The time required to reach this value is approximately $0.17s$, from which the time constant of the system is determined to be $\tau = 0.17s$.

The cutoff frequency is:

$$f_c = \frac{1}{2\pi\tau} \approx 0.936 \text{ Hz} \tag{5.33}$$

According to the Nyquist criterion,

$$f_{\text{sampling}} \geq 2f_c \geq 1.87 \text{ Hz} \Rightarrow \delta t \leq 0.534\text{s} \tag{5.34}$$

Therefore, the sampling interval is chosen as $\delta t = 0.01s$. For an observation duration of $T = 0.5s$, the sample size N is determined from:

$$\delta t \cdot N = T \Rightarrow N = 50 \tag{5.35}$$

The recorded input–output data were processed using MATLAB's System Identification Toolbox. A first-order transfer function was found to provide an adequate approximation of the motor dynamics within the operating range of interest. The identified transfer function is given by:

$$G(s) = \frac{33.0998}{s + 5.9748} \tag{5.36}$$

The identified model achieves a 97.46% fit with the experimental data, indicating that the dominant dynamic characteristics of the motor are modeled quite adequately.

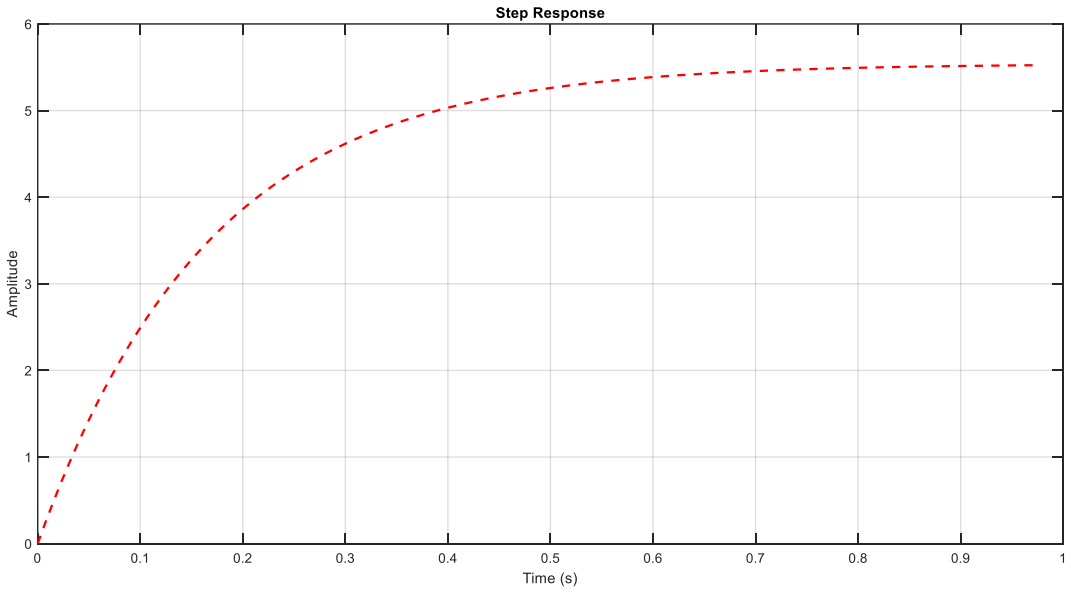


Figure 5.10. Step response of the identified motor transfer function

Key time-domain characteristics of the modeled motor response are summarized in Table 5.1. The absence of overshoot and the short settling time indicate a well-damped system, which is desirable for precise speed control in line-following applications.

Table 5.2. Step response characteristics of the motor

Parameter	Value
Rise time (s)	0.368
Settling time (s)	0.655
Peak value	5.5362
Overshoot (%)	0

5.5.2. Motor Open-Loop Stability Analysis

Before designing the feedback controller, the stability of the motor system in the open-loop configuration must be evaluated. The motor transfer function was identified as a first-order system:

$$G_m(s) = \frac{33.0998}{s + 5.9748}$$

The characteristic equation of the open-loop system is:

$$s + 5.9748 = 0 \Rightarrow s = -5.9748 \quad (5.37)$$

Evaluation: The root of the characteristic equation (system pole) lies in the left half of the complex plane ($s < 0$). Therefore, the motor system is inherently stable and satisfies the BIBO (Bounded-Input Bounded-Output) stability criterion. This inherent stability provides a favorable basis for the subsequent design of the feedback controller.

5.6. Motor speed controller design

The objective of the motor speed controller is to regulate the wheel speed accurately while ensuring smooth motion and robustness against disturbances. Although PID control is commonly employed, the derivative term is highly sensitive to encoder quantization noise and may amplify measurement errors. Moreover, the identified motor dynamics exhibit first-order behavior without significant overshoot.

For these reasons, a PI controller is selected. The proportional term improves transient response, while the integral term eliminates steady-state error caused by load variations and model uncertainties. This control structure offers a good balance between performance and implementation simplicity.

The design requirements are defined as follows:

- Overshoot: 0%
- Settling time: $T_s < 0.3s$
- Steady-state error: $e_{ss} \leq 5\%$

5.6.1. Analytical PI controller design

The identified transfer function of the motor is:

$$G_m(s) = \frac{33.0998}{s + 5.9748} \tag{5.38}$$

The PI controller is expressed as:

$$G_c(s) = K_P + \frac{K_I}{s} \tag{5.39}$$

The closed-loop transfer function can be written as

$$T(s) = \frac{33.0998(K_P s + K_I)}{s^2 + (5.9748 + 33.0998K_P)s + 33.0998K_I} \tag{5.40}$$

To achieve a non-overshooting response, a critically damped second-order system is assumed with damping ratio $\zeta = 1$. Based on the settling time requirement, the desired natural frequency is chosen as

$$\omega_n = \frac{4}{\zeta T_s} = 13.33 rad/s \tag{5.41}$$

Thus, the desired characteristic equation is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 26.66s + 177.78 \tag{5.42}$$

By equating the denominator of the closed-loop transfer function with the desired characteristic equation, the controller gains are obtained from

$$\begin{cases} 5.9748 + 33.0998K_P = 26.66 \\ 33.0998K_I = 177.78 \end{cases} \tag{5.43}$$

Solving the above equations yields

$$K_P = 0.62;\ K_I = 5.37$$

5.6.2. Controller selection and validation

Although the analytically derived gains satisfy the design requirements, they result in relatively high control effort and increased sensitivity to measurement noise. To improve robustness and ensure reliable real-time implementation, the final controller parameters were refined using MATLAB PID Tuning Tool.

The selected controller gains are

$$K_P = 0.3975, K_I = 2.3751$$

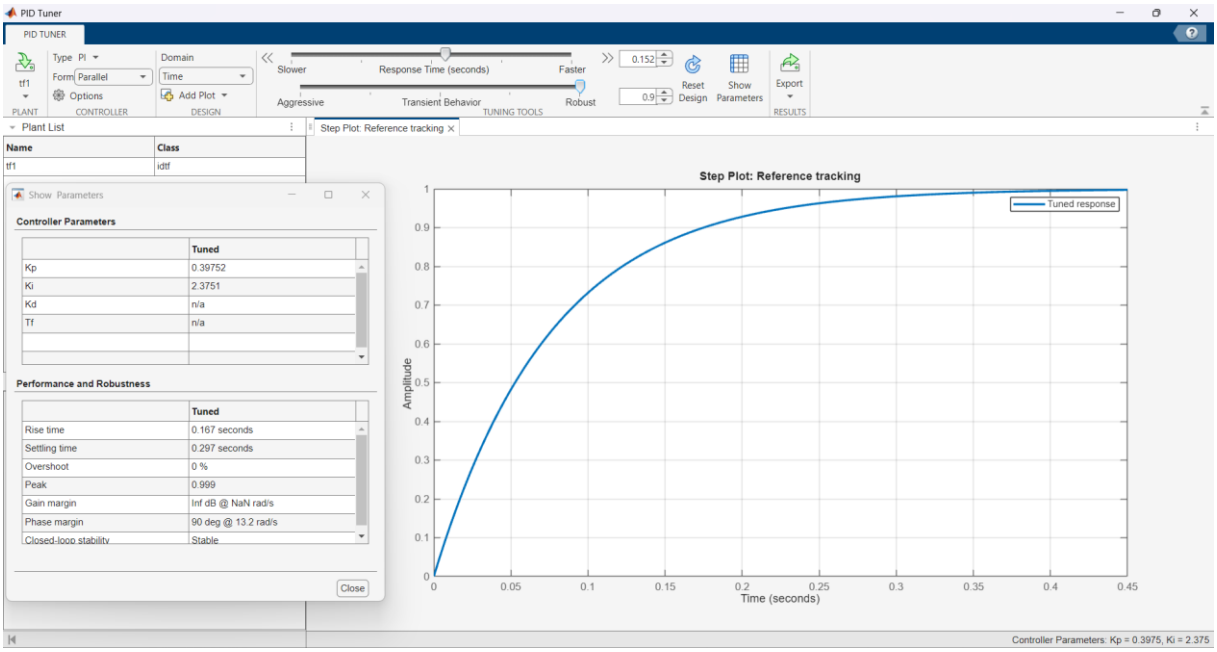


Figure 5.11. PI controller tuning results

Simulation results, shown in Figure 5.11, confirm that the tuned controller satisfies the performance specifications while maintaining smooth and stable motor speed control.

After tuning the controller parameters through simulation and selecting the optimal set of gains, the PI controller was implemented on the microcontroller for experimental validation using the actual motor hardware. A step reference speed of $717 RPM$ was applied to the system, while the actual motor speed was measured using the encoder.

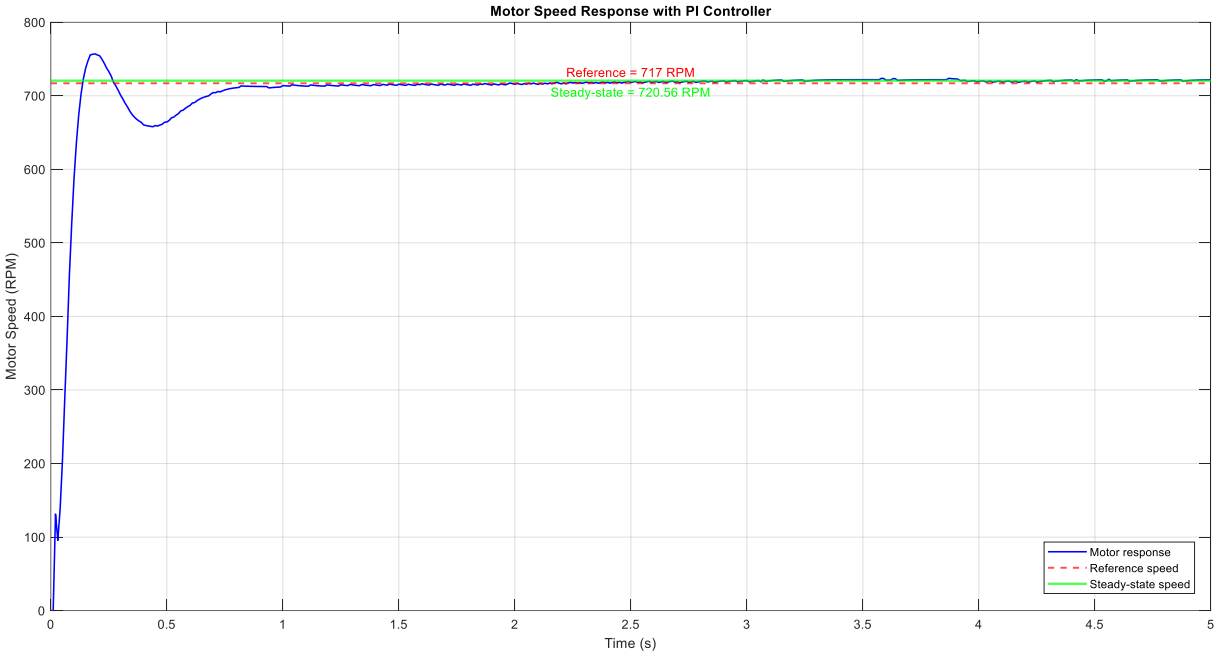


Figure 5.12. Experimental motor speed response

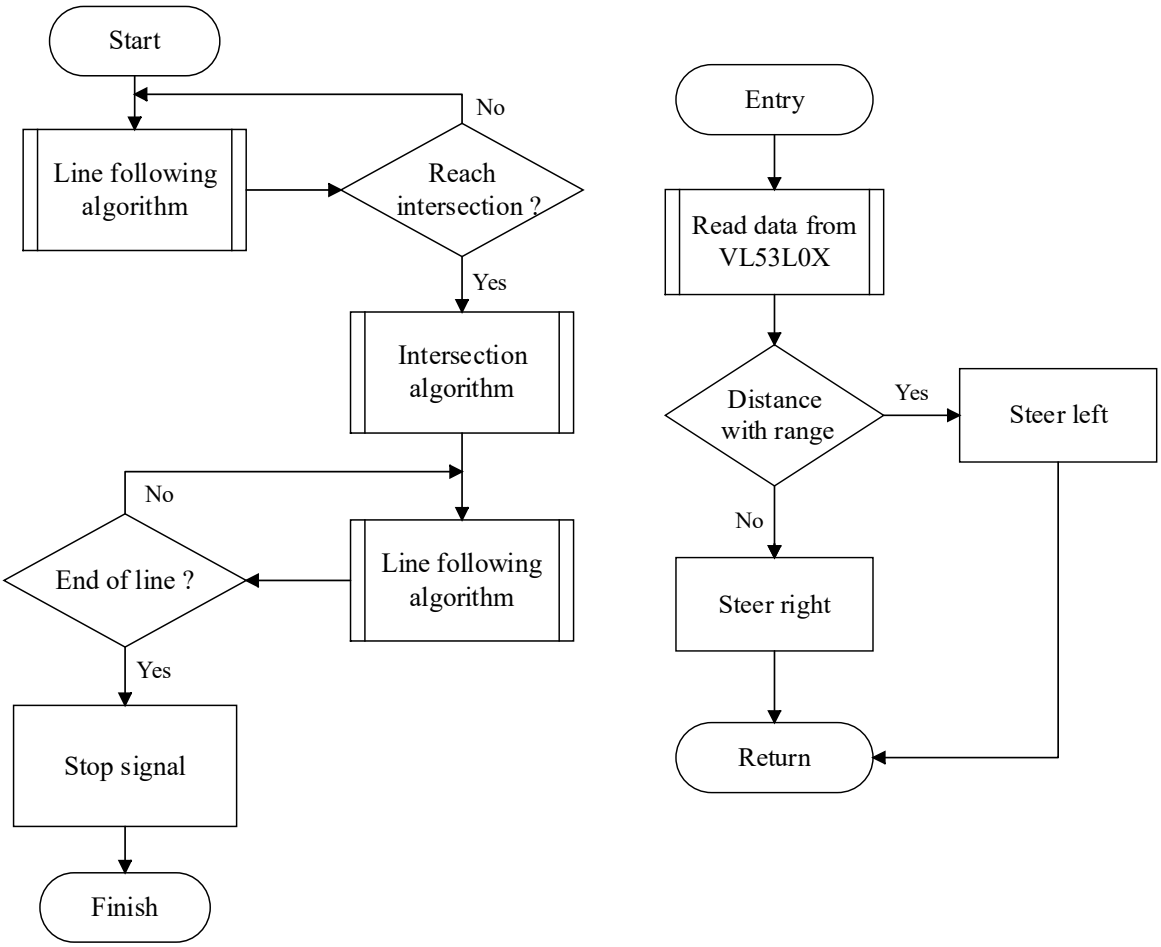
Figure 5.12 presents the experimental motor speed response under PI control. The motor speed initially exhibits a rapid increase, reaching a peak of approximately $750 RPM$,

corresponding to a moderate overshoot of about 4.6% relative to the reference. The speed then decreases and exhibits a small oscillatory transient before gradually converging toward the reference value.

In the steady-state region, the measured motor speed is approximately $720.56RPM$, which is very close to the reference value of $717RPM$. This corresponds to a steady-state error of approximately $3.56RPM$, or 0.50%. The small residual deviation can be attributed to encoder measurement noise, motor-load variations, and practical limitations of the hardware implementation.

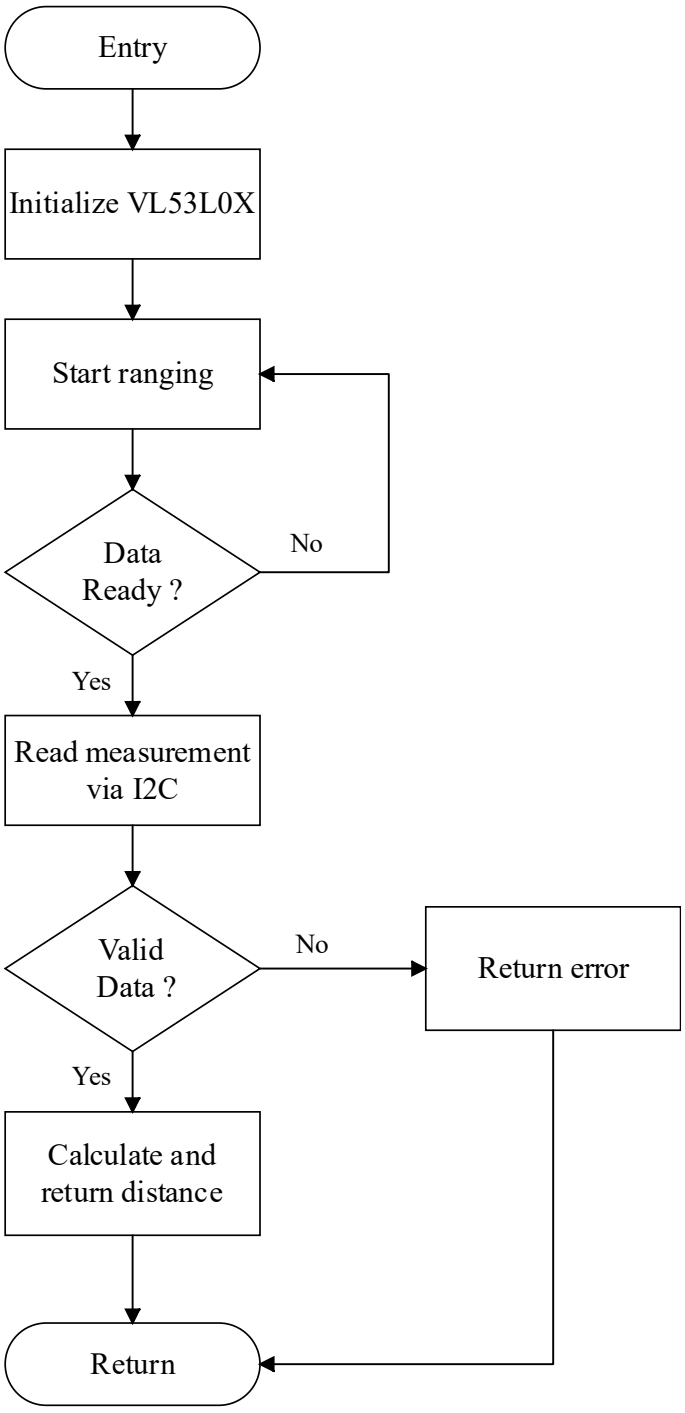
Overall, the experimental response demonstrates that the designed PI controller provides stable speed regulation with a small overshoot and low steady-state error. The motor reaches and maintains a speed close to the desired reference, confirming the effectiveness of the controller parameters obtained from the simulation and their suitability for the actual motor system.

CHAPTER 6. CONTROL ALGORITHM



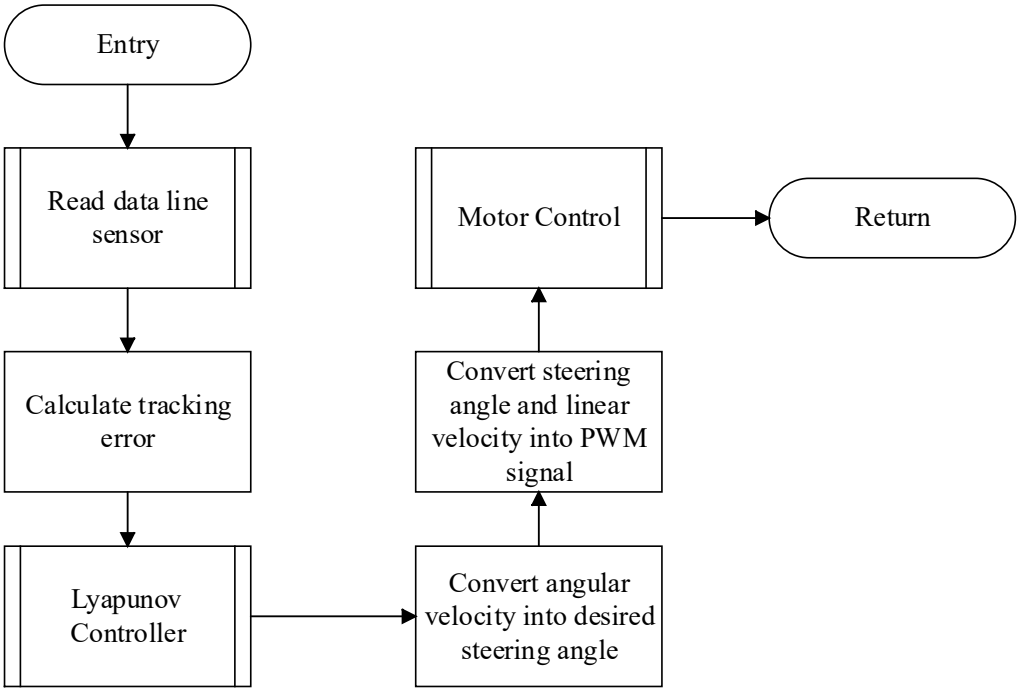
The overall operating algorithm is shown in Figure 6.1. After starting, the robot follows the predefined line while continuously checking for intersections. When an intersection is detected, the intersection algorithm is executed to select and perform the appropriate route. The robot then resumes line following. This process continues until the end of the line is reached, at which point a stop signal is generated and the robot finishes its operation.

The intersection algorithm is executed when the robot reaches an intersection. The obstacle information is determined in advance using the VL53L0X sensor before the robot starts moving. Based on the detected distance, the system determines the appropriate steering direction. The robot then performs the corresponding steering maneuver and resumes the line-following algorithm to continue along the selected route.



The VL53L0X distance measurement algorithm begins by initializing the sensor. The system then starts a ranging measurement and checks whether the measurement data is ready. If the data is not ready, the system waits and continues checking until a valid measurement becomes available. Once the data is ready, the measurement result is read from the VL53L0X through the I2C interface. The system then checks the validity of the received data. If the measurement

is invalid, an error is returned. Otherwise, the measured value is processed and the distance is returned to the main control algorithm.



The line-following algorithm begins by acquiring data from the line sensor array. Based on the sensor data, the tracking error is calculated to determine the deviation of the robot from the desired path. The tracking error is then processed by the Lyapunov controller to generate the required linear and angular velocity commands. The angular velocity is converted into the desired steering angle, while the steering angle and linear velocity are converted into PWM signals for actuator control. The motor control system then applies the corresponding commands to the actuators, allowing the robot to continuously follow the predefined path.

CHAPTER 7. EXPERIMENT AND RESULTS

CHAPTER 8. CONCLUSION

TÀI LIỆU THAM KHẢO

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